

[503] (CONTINUED)

On the Vertices of Cones which satisfy Six Conditions, particularly Six Conditions of Contact with a given Surface.

[Continuation from earlier pages, book pp. 117–134.]

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Factor opened upon by $\Delta = XI_0 + YI_4 + ZI_5 + WI_6$, reduces itself for $z = 0, w = 0$ to a multiple of W . Considering the factor in the form of a determinant, the result of the operation is

$$\begin{vmatrix} X, & Y, & Z, & W \\ \sqrt{p\alpha\beta\gamma} \cdot p\delta\delta, & x_1, & y_1, & z_1 \\ \sqrt{p\alpha\beta\gamma} \cdot p\delta\delta, & x_2, & y_2, & z_2 \\ \sqrt{p\alpha\beta\gamma} \cdot p\delta\delta, & x_3, & y_3, & z_3 \end{vmatrix} = \begin{vmatrix} . & . & . & . \\ \Delta\sqrt{p\alpha \cdot p\alpha\beta}, & a_1, & b_1, & c_1 \\ \Delta\sqrt{p\alpha \cdot p\alpha\beta}, & a_2, & b_2, & c_2 \\ \Delta\sqrt{p\alpha \cdot p\alpha\beta}, & a_3, & b_3, & c_3 \end{vmatrix},$$

the first term in

$$\begin{vmatrix} . & X & Y & Z & W \\ . & \sqrt{p\alpha \cdot p\alpha\beta} & a_1 & b_1 & c_1 \\ . & \sqrt{p\alpha \cdot p\alpha\beta} & a_2 & b_2 & c_2 \\ . & \sqrt{p\alpha \cdot p\alpha\beta} & a_3 & b_3 & c_3 \end{vmatrix}$$

where the first column may be replaced by

$$-\Sigma\sqrt{I_0I_3},$$

and the term in question then becomes

$$\sqrt{p\delta \cdot p\delta\delta} \cdot W(-x\sqrt{I_0I_3} + \sqrt{p\alpha \cdot p\alpha\beta}) \cdot \text{etc.}$$

if for shortness

$$|x_1\sqrt{I_1I_2} + W(-x\sqrt{I_0I_3} + \sqrt{p\alpha \cdot p\alpha\beta}) \cdot \text{etc.}| = \begin{vmatrix} x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \\ x_3, & y_3, & z_3 \end{vmatrix} \cdot \text{abc.}$$

As regards the second term, we have

$$\Delta\sqrt{p\alpha \cdot p\alpha\beta} = \frac{p\alpha \cdot \Delta p\alpha\beta + p\alpha\beta \cdot \Delta p\alpha}{2\sqrt{p\alpha \cdot p\alpha\beta}}.$$

But

$$\begin{aligned} p\alpha\alpha &= x(-p_1a_1) + y(-f_1a_2) + z(-g_1a_3) + w(h_1 - a_4) = h_1(x - a_1) - a_2b_1(x - a_2) - a_3c_1(x - a_3), \\ &= -p_1(x - a_1) - f_1(y - a_2) - g_1(z - a_3) - h_1(w - a_4), \end{aligned}$$

and thence

$$\Delta p\alpha\alpha = a_1(p_1a_1 + f_1a_2 + g_1a_3 + h_1a_4) - p_1x_1 + f_1y_1 + g_1z_1 + h_1w_1.$$

[Sadleirian $abc\ell'c$].

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with the like formula for $\Delta p\delta\beta$; hence

$$\frac{I_1\Delta p\delta\beta + I_2\Delta p\alpha\alpha}{2\sqrt{I_1I_2}\sqrt{p\delta\delta \cdot p\alpha\beta}} = Ax_1 + By_1 + Cz_1,$$

where

$$\begin{aligned} A &= \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} \{f_2I_1(p_1a_1 - h_1W) + f_1I_2(p_2a_2 - h_2W)\}, \\ B &= \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} \{g_2I_1(g_1a_1 - k_1W) + g_1I_2(g_2a_2 - k_2W)\}, \\ C &= \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} \{h_2I_1(h_1a_1 - l_1W) + h_1I_2(h_2a_2 - l_2W)\}. \end{aligned}$$

The term in question is thus

$$\frac{Ax_1 + By_1 + Cz_1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} = \begin{vmatrix} \cdot & x & y & \cdot \\ Ax_1 + By_1 + Cz_1, & x_1, & y_1, & z_1 \\ Ax_2 + By_2 + Cz_2, & x_2, & y_2, & z_2 \\ Ax_3 + By_3 + Cz_3, & x_3, & y_3, & z_3 \end{vmatrix}$$

viz. replacing the first column by

$$\begin{aligned} & -Ax - By \\ & \Delta\sqrt{p\delta\beta \cdot p\delta\delta} - Ax_1 - By_1 - Cz_1, \\ & = (Ax + By)x_1 \cdot abc; \end{aligned}$$

and we have

$$\begin{aligned} Ax + By &= \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} \{f_2I_1 \cdot p_1x - f_1g_1(z - a_3) + g_1g_2(g_1a_2 - h_2W)\} + \frac{1}{2} \{h_2I_1(h_2x - h_2a_3 - l_2W)\}W, \\ &= \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} \{I_1p\delta\delta_2 - I_2HW\} \end{aligned}$$

if for shortness

$$M = (-p_2a_1 + f_2a_2)(g_2z + h_2y) + (-p_2a_1 + f_2a_2)(g_2z + h_2y);$$

viz. the whole term is

$$Wx \left(-\sqrt{I_0I_3} \frac{H}{\sqrt{I_1I_2}} + \frac{H}{\sqrt{I_3I_4}} \right) \cdot abc;$$

Hence the first and second terms together are

$$= W \left(-x\sqrt{I_0I_3} + \sqrt{p\alpha \cdot p\alpha\beta} - \frac{H}{\sqrt{I_2I_5}} \right) \cdot abc;$$

viz. this is a multiple of W , which was the theorem to be proved. [*Sadleirian abcℰc*].

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Surface chiefly.

32. The equation is

$$\text{Norm} \left(\sqrt{p\alpha\alpha}, \sqrt{p\delta\beta}, \sqrt{p\gamma\gamma}, \sqrt{p\delta\delta} \right) = 0,$$

where the norm is a product of 16 factors, each of the order $\frac{1}{2}$. As before, $p\alpha\alpha = 0$ is the equation of the plane through the point α and the line α ; viz. $p\alpha\alpha$ has the value already mentioned.

33. *Investigation.* In the projection, the equation of the conic touching the projection of the lines $\mathfrak{a}, \mathfrak{B}, \eta$ is

$$A\sqrt{p\alpha\beta} \cdot Q\sqrt{P} + R\sqrt{B} + S\sqrt{Q} + R\sqrt{Q} + R\sqrt{S} = C\sqrt{P} + S\sqrt{Q} + R\sqrt{B} = 0;$$

and to make this pass through the projection of the line δ , we must write herein $X : Y : Z = p_3 : g_3 : c_3$. As before, we have

$$\begin{aligned} P_3X + Q_3Y + R_3Z &= m_3(x \cdot P_3 + y \cdot Q_3 + z \cdot R_3), \\ &= (a_1b_2 - a_2b_1)(\text{etc.}), \\ &= (a_1P_3 + y_2Q_3 + z_3R_3)p\alpha\alpha_3, \\ &= -abc \cdot p\alpha\alpha_3; \end{aligned}$$

and so for the other terms; the equation thus is

$$A\sqrt{p\alpha\alpha} + B\sqrt{p\alpha\beta} + C\sqrt{p\gamma\gamma} = 0;$$

or freeing the like equations in regard to the points $\mathfrak{a}, \mathfrak{B}, \gamma$ respectively, and eliminating we have a determinant $= 0$, and then, taking the norm, we obtain the above-written equation of the surface.

34. *Sub-projection.* The equation shows that the surface chiefly shows that (1) The point α is 8-conical: in fact, by the point in question we have $p\alpha\alpha = 0$, $p\alpha\beta = 0$; each factor is O , and the norm is O^8 . (2) The line $\mathfrak{a}$ is 4-tuple. To see this, observe that the sixteen factors are obtained by attributing as pleasure the signs $+$, $-$ to the radicals $\sqrt{p\delta\beta}, \sqrt{p\delta\beta}, \sqrt{p\delta\gamma}, \sqrt{p\delta\delta}$; hence there are four factors in which $\sqrt{p\delta\beta}, \sqrt{p\delta\beta}$ have determinate signs, but in which we attribute to the radicals $\sqrt{p\delta\gamma}, \sqrt{p\delta\delta}$ the signs $+$ or $-$ at pleasure. It is to be shown that the four factors each vanish for a point on the line of; that is, on writing therein for $\mathfrak{a}, y, z, w$, the values $a_1, wa_1, -wa_1, b_1$. But we then have $\Sigma p_1b\beta_1 = p\alpha\beta$. And the whole expression thus is [*Sadleirian* $abc\mathcal{E}c$].

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have, as before, $p\alpha\alpha = -z \cdot abc$ and $p\delta\alpha = a \cdot abc$, with the like formula with $\beta\delta$ and γ in place of α . The factor thus becomes

$$\sqrt{-z \cdot abc} \begin{pmatrix} \sqrt{abc}, \sqrt{abc}, \sqrt{p\delta\beta} \\ \sqrt{abc}, \sqrt{abc}, \sqrt{p\delta\beta} \end{pmatrix}$$

which vanishes, being $= 0$, and the norm is thus $= 0$, viz. the line is 4-tuple. (3) The line δ is 4-tuple: in fact, writing $w = 0$, $w = 0$ for the equations of the line, we have $A_0 = 0$, $b_0 = 0$, $b_2 = 0$, and $g_1g_2 - g_2g_1 = 0$, or say $p\delta\beta = b_1g_3$, $p\delta\beta = b_3g_1$. Hence, writing

$$I = (g - bc)g - (f + bd)g,$$

viz. $I_1 = (g_2 - b_3)g_1 = b_1g_2g_3$, etc. for the conditions of the line δ , etc. in the equations of the line δ , etc. for the conditions $p\delta\delta = b_3g_1$, $p\delta\beta = g_2g_1$, etc. and $a_1a_3 - a_2a_2 = 0$, or any abc , $b_1b_2 = b_2b_1$, etc. Hence, writing

$$I = (-g - bc)g - (f + bd)g.$$

The factor thus is

$$\begin{vmatrix} \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \end{vmatrix}$$

which vanishes, being $= 0$; there are four such factors, or the norm is O^4 ; whence the line is 4-tuple. (4) The line $a\beta$ is a 4-tuple line. To prove it, take as before $\alpha = w = 0$ as the equation of the plane $\alpha\beta$, and $\xi = 0$, $w = 0$ for the equations of the line in question. We have $h_0 = 0$, $g_1 = 0$, $c_2 = 0$, $b_2 = 0$, $f_2 = 0$, $b_2 = 0$, whence $(\Sigma a_2b_3, c_3, w)$ as before, $p\alpha\alpha = f_3$, $p\alpha\beta = f_2$, $p\alpha\gamma = -p_3z - h_3W$, etc., and similarly $p\alpha\beta, p\delta\beta, p\delta\gamma$ have I_3 etc., $I_4 = abc$. Also $I_5 = abc$, the factor thus is

$$\begin{vmatrix} \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{p\delta\beta}, & \sqrt{p\delta\beta} \end{vmatrix}$$

which vanishes, being $= 0$; and there are four such factors, or the norm is O^4 ; viz. the line is 4-tuple. (5) The line $\alpha\beta$ is a 4-tuple line. To prove it, take as before $z = 0$ as the equation of the planes $\alpha\beta$, and $\xi = 0$, $w = 0$ for the equations of the line in question. We have $p_1 = 0$, $f_3 = 0$, $g_3 = 0$, $b_2 = 0$, and similarly $p\delta\delta, p\delta\delta, p\delta\delta = I_3\alpha_1c_3$. Hence, writing for shortness $I = \mathfrak{C}p - g_3p - h_3p$, the factor is

$$\begin{vmatrix} \sqrt{I_3I_2}, & \sqrt{I_3I_3}, & \sqrt{abc} \\ \sqrt{I_3I_2}, & \sqrt{I_3I_3}, & \sqrt{abc} \\ \sqrt{p\gamma} \cdot \sqrt{p\delta}, & \sqrt{p\gamma}, & \sqrt{p\delta} \end{vmatrix}$$

which vanishes, being $= 0$ and there are four such factors, obtained by giving to the radicals the signs $+$, $-$ at pleasure; hence the norm is $[Saddleirian\ abc\mathfrak{C}c]$.

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Surface chiefly.

35. The equation is

$$\text{Norm} \left(\sqrt{p\alpha\alpha}, \sqrt{p\alpha\beta}, \sqrt{p\beta\gamma}, \sqrt{p\beta\delta}, \sqrt{p\delta\beta}, \sqrt{p\delta\beta} \right) = 0,$$

where the norm is the product of 8 factors each of the order $\frac{1}{2}$. As before, $p\alpha\alpha = 0$ is the equation of the plane through the point α and the line $\mathbf{a}$; viz. $p\alpha\alpha$ has the value previously mentioned. And of course $p\beta\beta = 0$ is the equation of the plane through the point β and the line $\mathbf{a}$, etc.

36. *Investigation.* In the projection, taking ξ, η, ζ as current line-coordinates, the equation of the conic passing through the projections of the points α, β is

$$\sqrt{(p\alpha\beta + a_1\alpha P + b_2a_1P_1)} + \sqrt{(p\alpha\beta + a_3\alpha P + b_3a_2P_2)} \cdot A\xi + B\eta + C\zeta = 0,$$

where A, B, C are arbitrary constants. To make this touch the projection of the line a , we must write $\xi : \eta = -p_1 : h_1 - h_1$; and similarly,

$$\begin{aligned} p\delta\xi + p\delta\eta + p\delta\zeta &= a_1p\beta\beta\xi + b_1c_1, \\ &= a_1p\delta\beta\xi + b_2c_2, \end{aligned}$$

and similarly

$$p\delta\xi + p\delta\eta + p\delta\zeta = -a_1p\beta\beta\xi.$$

Hence the equation is

$$a\sqrt{p\alpha\beta \cdot p\alpha\beta} + \beta P_2 + \gamma Q_3 + \delta R_3 = 0;$$

or freeing the like equations for the lines $\mathbf{a}, \eta, \delta$ respectively, and eliminating, we have

$$\begin{vmatrix} \sqrt{p\alpha\alpha \cdot p\beta\alpha}, & P_1, & Q_1, & R_1 \\ \sqrt{p\alpha\alpha \cdot p\beta\alpha}, & P_2, & Q_2, & R_2 \\ \sqrt{p\alpha\alpha \cdot p\beta\alpha}, & P_3, & Q_3, & R_3 \end{vmatrix} = 0;$$

which, throwing out a factor a , becomes

$$\sqrt{p\alpha\alpha \cdot p\beta\beta} + \sqrt{p\alpha\alpha \cdot p\beta\beta} + \sqrt{p\gamma\gamma \cdot p\delta\delta} + \sqrt{p\delta\delta \cdot p\delta\delta} \cdot p\beta_3 = 0;$$

or, taking the norm, we have the above-written equation.

37. *Singularities.* The equation shows that (1) The point α is 8-conical point; in fact, for the point in question we have $p\alpha\alpha = 0$, $p\alpha\beta = 0$; and $p\alpha\alpha = 0$; each factor is $= O$, and the norm is $= O^8$. [*Sadleirian abcℰc.*]C. VIII.

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(2) The line $\alpha\beta$ is a 2-tuple line. To prove this, we have for the coordinates of a point on the line in question $\alpha_1 + r\alpha_2, \eta_2 + r\eta_2, \&c.$; the values of $p\alpha\alpha, p\beta\beta$ become as below: r does not enter, but similarly for $p\delta\beta, p\delta\delta, \&c.$, so that, omitting the constant factor $\sqrt{r} = \text{etc.}$, the result of the operation is

viz. the value of the factor is $|a\sqrt{(fgh)} + g\sqrt{(bdh)} + h\sqrt{(dcg)} + h\sqrt{(cbg)}|$, is the determinant

$$\begin{vmatrix} f & b & g & h \\ \cdot & \cdot & \cdot & \cdot \end{vmatrix},$$

the suffixes in the four lines being $\mathfrak{a}, \mathfrak{B}, \gamma$ respectively. Collecting, this is

$$\begin{aligned} & (\quad + cf g f g - bc f g s + f g h f g) \\ & (-ce f g h + ce f g s + pc f s h p \cdot bc) \quad + \xi z + b \xi \alpha) \\ & (+bc f g h + bc f g s + b k(b p bc)) \quad + \xi \alpha + b \xi \alpha) \\ & (-bc f g h + bc f g s + b(b a) p bc) \quad + \xi \alpha + b \xi \alpha) \\ & + bc g h \{ w(ax + by + cs) + x \} \\ & + \pi(ag \xi \eta + bg \xi z) + ag - dg(ax + by + cs - x) \\ & + \pi(\alpha \xi \eta + b \eta \xi) (bx + ay + cz)(ax + by + cs + a bcc) \} \end{aligned}$$

or, what is the same thing,

$$AP + BQ + CR + DS = 0,$$

[Sadleirian abc'c.]

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where

$$\begin{aligned} A &= [-bgh + abfh - bfgs + fcbcw], \\ B &= [-bgh + bfh + bgfh + abfsw], \\ C &= [b bgh - b fgh + b gfgs + bcdsw], \\ D &= [-bgh - bcfg - bcfgs + (bcdh + cabsf + abfh)w], \\ P &= x(-by + gz + bwx), \\ Q &= (-bx + bz + hw), \\ R &= (gx + hy + aw), \\ S &= (bwx + aby + c wz - bx); \end{aligned}$$

the right-hand factors vanishing for the values $\alpha_1 + r\alpha_2, \eta_2 + r\eta_2$, &c., of the coordinates.

38. It thus appears so far that the factor is $= O$; it is, in fact, $= O^2$, viz. we can show that, operating upon it with

$$\Delta = XI_0 + YI_4 + ZI_5 + WI_6,$$

the value of $\Delta p\alpha\alpha$ is

$$\Delta\sqrt{p\alpha \cdot p\beta\delta} = \frac{p\alpha \cdot \Delta p\beta\delta + p\beta\delta \cdot \Delta p\alpha}{2\sqrt{p\alpha \cdot p\beta\delta}} + \sqrt{p\alpha \cdot p\beta\delta} \cdot \Delta\sqrt{p\alpha \cdot p\beta\delta},$$

where $\Delta\sqrt{p\alpha \cdot p\beta\delta}$ is what $p\alpha$ becomes on writing therein (X, Y, Z, W) in place of (x, y, z, w) . Writing as before, for α, β, γ the values $a_1 + r a_2$, &c., we have $p\alpha\alpha = -r \cdot abc$, $p\beta\beta = -r \cdot abc$, and putting for shortness

$$-\Pi\Theta = -r\Sigma h\Sigma\xi\Sigma + \cdot\&c.,$$

the expression in question, divided by $\sqrt{-r}$, is

$$\begin{aligned} &= -B\sqrt{abc \cdot \Delta\Pi p\delta\beta} + \text{etc.} \\ &= (B\alpha \cdot p\beta\delta)\sqrt{-r}. \end{aligned}$$

where, denoting the determinants

$$\begin{vmatrix} X & Y & Z & W \\ \cdot & \cdot & \cdot & \cdot \end{vmatrix}$$

by (d, b, c, f, g, h) , we have

$$hh = a\Sigma_1 + b c_1 + c c_1 - b c_1 + r\Sigma_1 + b c_1,$$

which is an identity in regard to (x, y, z, w) ; and the expression is

$$\Delta p\delta\beta = \Delta abc \cdot p\delta\beta, \quad \Delta p\delta\beta = abc \cdot p\delta\beta + r \cdot \Delta\beta\delta\beta,$$

But also $\Delta p\delta\beta = \Delta abc \cdot p\delta\beta - r[(-1 - c) + (-1 - c) + 2cD\beta] + Bb$, that is,

$$\Delta p\delta\beta = abc(P - r^2(b - c)^2) + DD.$$

[Sadleirian abcℰc.]

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where P, Q, E, R denote $by - gx + ax, \&c.$, and where, finally, x, y, z, w are to be replaced by $\alpha_1 + r\alpha_2, \&c.$ Since for these values P, Q, R, S vanish, the expression becomes

$$= -B\alpha(\Delta\Sigma P + \Delta\beta Q + \Delta\gamma R + \Delta\delta S) + AP + YQ + ZR + SW = 0;$$

that is

$$= A(\Sigma P + \beta\Delta Q + \gamma\Delta R) + AP + BQ + CR + DS = 0,$$

and we have, in fact, $P = \alpha\Delta P\beta R = \alpha\Delta P + Q\Delta\Sigma R\beta = D(P + 2\alpha\Delta\Sigma R)$,

$$P + VQ + VR = ba_1, \quad \&c. \text{ for the suffix.}$$

$x, y, z, w = -\alpha_1, ba_2, \dots, \eta_2, ba_2, ba_2, \dots$, where, finally, x, y, z, w are to be replaced by $a_1ba_2 + a, \&c.$ $x, y, z, w = 1, m, n; m, p; n, p; m, p; n, p, m, p, m, p, m$; then, for instance,

$$H = a, X + b, Y + c, Z,$$

where

$$a, b, c = b, c - b, c \quad c, a - c, a; \quad Xy - Y\alpha;$$

and thence

$$H = X, Y, Z; \quad XYy - YZx;$$

$$H = \begin{vmatrix} x & y & z \\ b & b & b \end{vmatrix}$$

$$= 2ax(aX + bY + cZ)$$

$$= 2\nu a\beta XY.$$

and similarly the factor is $= 0$; the factor is thus $= O^2$; there is only one such factor, and the line ab is double. (2) The line b is an 8-tuple line: in fact, for a point on the line we have $p\alpha\alpha = O, p\delta\beta = O, p\delta\beta = O, p\delta\beta = O, p\delta\delta = O$, and the factor vanishes, being $= O$. Each of the factors is O , and the norm is $= O^8$. 39. (3) The line $[a\delta, x, \beta, b]$ is a double line. To prove this, observe that this for a point on this line we have $p\alpha\beta = O$. Taking as before $z = 0, w = 0$ as the expressions of the plane $\alpha\beta$, and as in $\xi = 0, w = 0$ for those of the line $\alpha\beta$, we have $h_2 = 0, a_3 = 0, p\beta\beta = 0$, and $p\beta = h\delta = h\delta$, etc. in the expressions of $p\beta\beta\delta$, this, may be written

$$\Sigma[(p - h\delta - (f + b\delta)y] h\delta\delta \cdot [k\delta\delta - (a + b\delta)y - (bk + bd)x] + (b\delta b)z];$$

where observe that Σ denotes a sum of three terms of the form

$$b_1.[p_3.\delta_3 z]\beta\beta_2 = h\delta;$$

[Sadleirian $abc\mathcal{E}c.$]

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Adding thereto a fourth term $-1, abp$, the value of this sum becomes $= ablp^2$, or the sum of the three terms is $= ablp^2 + 1, abp$; where the symbols represent determinants. But in each case the determinant $ab\ b\delta = 0, b\delta = 0$, since the column b_1, b_2, b_3 , the terms of which are each $= 0$. Say $ba_1b_3a_2 = pgh - p_3gh$, where in δ, h the suffixes are at a, δ, b , and in each case appears under a $\sqrt{}$, we have $\Sigma\delta_1, abp = pgh\delta$. And the whole expression thus is

$$\begin{aligned} &= a^2(pgh - hs\delta gh) \\ &+ p(pgh)\delta(-hs\delta gh) + a\delta gh(hs\delta gh - hs\delta gh) \\ &+ pf(pgh)\delta - hs\delta gh) \\ &+ pgh(-hs\delta gh) - agh(-hs\delta gh) \\ &+ pgh(-hs\delta s\delta) - p\delta - hs\delta gh) \\ &+ q^2(-hs\delta - p\delta gh) \end{aligned}$$

where $pgh\delta$ denotes the determinant $p_3\ g, h_3, f$, the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta$, in the forms respectively, and so on; the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta$ are each as $\delta(\delta + \delta - \delta)$ of a sign of δ at pleasure), and the norm is thus $= O^2$.

40. But the line is bicuspid, each sheet of the surface touching along the line in question the hyperboloid $p\beta\beta$. To prove this, write

$$\Delta = XI_a + YI_b + ZI_c + WI_d;$$

we have for the hyperboloid, writing $z = 0, w = 0$,

$$\Delta p\beta\beta = a\delta(z = h)\beta, \quad p\delta\beta = c\delta(z = h\delta)y,$$

and so

$$\Delta(\sqrt{p\delta\beta}, p\delta\beta, p\delta\beta, p\delta\delta, p\delta\beta, p\delta\beta, p\delta\delta);$$

each contains the factor $\Delta p\delta\beta$; or, what is the same thing, that

$$\Delta \sqrt{p\delta\beta, p\delta\beta},$$

contains the factor in question, Σ denoting the sum of the first three terms of the original expression. The value is

$$\sqrt{\frac{p\alpha \cdot Ph\delta + p\delta\beta \cdot p\delta}{2\sqrt{p\alpha \cdot p\beta\delta}}} p\delta\beta + \sqrt{p\alpha \cdot p\beta\delta} \cdot \Delta p\beta\delta;$$

$$+ p\delta\beta\delta \cdot \beta\delta(\delta + \delta) - p\delta\beta\delta\delta \cdot p\delta\beta\delta = 0, \text{ [Sadleirian } abc\ell c.]$$

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where we have

$$\begin{aligned}x_1 \Pi ba &= b\delta a, \\y_1 \Pi ba &= c\delta b.\end{aligned}$$

Also

$$\begin{aligned}a_1(a_1, [a_1 - Wa_2] + p_a(a_3 - Xa_4 - Wa_2) + a_5[(Xp_3 - z_3)p_a + b(Xp_4 - Wa_2)] \\+ a_5[(Xp_5 - z_3)p_a + b(Xp_4 - Wa_2)] + Y(Xp_5 - z_3)(Xp_6 - Wa_2) \\+ \Sigma(b_1, a_2b_3) - Zp_a - Wa_2) \\+ (a_1 - a_2 - Xa_2 - Xa_2), \\+ 2b_3a_2[X(a_3 - a_2)] + Y(bXp_5 - z_3)p_a + Y(bXp_5 + (b\delta - Wa_2)) = 0.\end{aligned}$$

Also

$$\begin{aligned}a_1ba &= a_3(z = z_a, + (a_3 + a_4)ba + (z_3 + b\delta)z), \\bab &= a^2p^2 + ap_b(a + b\delta)b + \delta b \\&\quad + Y[2a(a_3 - a_4) + bb a_5z] \\ \Delta bab &= X[2p_a + b\delta z] \\&\quad + Z[2a(a_3 - a_4) + (b - a)b - z], \\ \Delta abab &= W[2a(a_2 + p_a + b\delta a + b\delta y)\end{aligned}$$

41. The whole expression is a linear function of X, Y, Z, W , and it is easy to see *a priori*, or to verify, that the coefficients of X, Y , each of these vanish. The coefficient of Z is

$$\begin{aligned}&= \Sigma[a_1(a_2, ba_3) - (a_3, ba_2) + a_5(b_2, ba_4, ba_3) + 2ba_2(b_3, ba_2)p\delta\beta \\&\quad - \Sigma b_3(ba_2a_3, + b_3, ba_5) + ba_4 + 2a_5(ba_2 + ba_3, ba_5) + (b_3, ba_5) a_3b]\end{aligned}$$

with a like expression for the coefficient of W . The foregoing expression may be written

$$\begin{aligned}(b_a a_1 + a_5 b_3) \Sigma[ba_3 + a(p_a + b\delta z)p_a + b\delta y] &= b\delta a[b\delta a + a(p_a + b\delta z)p_a + b\delta y] \\&\quad + \Sigma b_3[ba(a + b\delta + b\delta y) + 2b_3p_a + b] \\&\quad + 2b_3ba(b_2, ba(a + b\delta + b\delta y + b\alpha y) - (aa(b\delta - g\delta))) \\&\quad + \Sigma b_3ba(b_2 - p_a)(p_a + b\delta + b\delta + b\delta + b\delta)\end{aligned}$$

The first sum is

$$\begin{aligned}a^2 \cdot p\alpha\alpha + ag(p\alpha\beta + p\delta\beta) + p\delta\beta\delta y + g\delta\beta, \\= -ag \cdot a fpha = -ag(fg\alpha + b)aa,\end{aligned}$$

where $afgh, \beta$ etc. denote determinants with the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma$. Similarly the second sum is

$$= -ag(g fpha, a)\beta a,$$

in the third sum is

$$= -ag(h fpha a)\beta\gamma;$$

and the fourth sum is

$$= -(a, p + b, p + c, p) a g fp.$$

[Sadleirian abc&c.]

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The whole coefficient of Z thus contains the factor $(cfy - x + bfy - z)$, and similarly it would appear that the whole coefficient of W contains the factor $(cfy - z + afy - bfx)$. the other factor being the same in each case; viz. the two terms together are

$$(-(a_1y_1 - ba_2ba_5)X[(afy + afy) + W(ay + afy - agy - afy)].$$

where the second factor is $\Delta p\beta\beta$, which is the required result. See *post*, No. 56 *et seq.*

42. (4) The line $[\delta, \beta, \beta]$ is an 8-tuple line: in fact, for any point of this line in question we have $p\delta\beta = 0, p\delta\beta = 0, p\delta\beta = 0, p\delta\beta = 0$; whence each factor is O , or the norm is O^8 .

I notice that the surface meets the quartic $p\beta\beta$ in

lines α, β, γ	each 8 times 24
„ (a, β)	16
„ $(\delta, a, \beta, \gamma)$	8
$24 \times 2 = 48$	

Surface octuple.

43. The equation is

$$(p\alpha\beta \cdot p\alpha\gamma \cdot p\alpha\delta \cdot p\alpha\delta + p\beta\gamma \cdot p\beta\delta \cdot p\gamma\beta \cdot p\delta\beta) = -bp\alpha\beta \cdot p\alpha\gamma \cdot p\alpha\delta \cdot p\beta\delta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\beta\alpha = 0;$$

or, what is the same thing,

$$(p\alpha\beta \cdot p\alpha\gamma \cdot p\alpha\delta \cdot p\alpha\delta - p\beta\gamma \cdot p\beta\delta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\beta\alpha = 0,$$

the equivalence of the two depending on the identity

$$p\alpha\beta \cdot p\beta\delta \cdot p\alpha\delta + p\delta\beta \cdot p\beta\alpha \cdot p\alpha\beta - p\delta\beta \cdot p\alpha\beta \cdot p\beta\delta = 0,$$

where, as before, $p\alpha\beta = 0$ is the equation of the plane through the lines $a, \beta \beta$; and $p\alpha\alpha = 0$ is the equation of the plane through the line and the point α ; viz. $p\alpha\beta$ &c., and *post*, No. 56, have the values already mentioned: $p\alpha\beta, p\beta\delta = 0$ are also mentioned in this cubic surface through the lines $a, \beta, \beta \beta$, and *vol, etpa. [Sadleirian abcℓc.]*

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44. *Investigation.* In the projection, using line-coordinates, the equation of the conic touching the five lines may be written

$$(P, Q, \Xi P)\beta = 0;$$

$$(P, Q, R)\beta = 0,$$

where the conical denotes a determinant: the last five lines of which are obtained by giving to (P, Q, R) the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta, \delta$ respectively. Writing herein the equation P, Q, R for the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta, \delta$ are at once transformed into

$$a\beta\delta \cdot p\beta\beta, a\beta\delta \cdot p\beta\delta, a\beta\delta \cdot p\delta\beta, a\beta\delta \cdot p\delta\beta, a\beta\delta \cdot p\delta\beta = 0,$$

or, what is the same thing,

$$p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0;$$

or say

$$p\alpha\beta \cdot p\beta\beta \cdot p\delta\beta + RX + Y(C\bar{A}' + CO'' + CO'') = 0;$$

$$-(p\beta\beta \cdot p\beta\beta \cdot p\delta\beta + A'\bar{Y} + B + C''\bar{Y})\Omega = 0;$$

where $p\beta\delta$, &c., signify as before; and

$$A\bar{a} + B + a + C\bar{a} = \bar{a}\bar{a};$$

$$\begin{vmatrix} P_1 & Q_1 & R_1 \\ \cdot & \cdot & \cdot \end{vmatrix},$$

$$\begin{vmatrix} P_2 & Q_2 & R_2 \\ \cdot & \cdot & \cdot \end{vmatrix}$$

and so for $A\bar{b}\beta + B + \beta + C\bar{c}\beta$ and $A'''\bar{Y} + B'' + C'''\bar{Y}$ being for (y, x) and (η, b) respectively. ξ, η so that the foregoing equation is

$$\begin{vmatrix} p\alpha\beta \cdot p\beta\beta & p\alpha\beta & p\delta\beta & p\delta\beta \\ A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix} + 4p\alpha\beta \cdot p\beta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \begin{vmatrix} p\delta\beta & p\delta\beta & p\delta\beta \\ A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix} = 0;$$

or in the equivalent form, when the first term we have + instead of -1, and in the second term the determinants are

$$\begin{vmatrix} P_1 & Q_1 & R_1 \\ A & B & C \end{vmatrix}, \quad \begin{vmatrix} P_2 & Q_2 & R_2 \\ A' & B' & C' \end{vmatrix},$$

46. To reduce this result, observe that we have: $A, B, C : bp - q, -q + p, bp - r, bpr, bp - r, -pX + fq - 0;$

$$by - gr + dr, \quad -br + r, \quad +fy - r;$$

[*Sadleirian abc'c.*]

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where, for convenience, I retain the unaccented and accented letters $(a_1, \dots)$, $(a', \dots)$ instead of those letters with the suffixes a and β respectively. Writing as before

$$\begin{aligned} L &= aag - aabhg + \dots \\ M &= aag - aabh + \dots \\ N &= aag - aabh + \dots \\ \Omega &= (gh - p')h + \dots; \end{aligned}$$

then

$$\begin{aligned} A &= L - Mw, \\ B &= L - Nw, \\ C &= \Omega - Nw, \end{aligned}$$

and similarly

$$\begin{aligned} A' &= L\Omega - Cw, \\ B' &= L\Omega - Nw, \\ C' &= \Omega - Nw, \end{aligned}$$

where for L', M', M', Ω' we have $(a', \dots)$ for $(a, \dots)$. Hence

$$BC' - CB' = w, \begin{vmatrix} \cdot & \cdot & \cdot \\ M, & N, & \Omega \\ M', & N', & \Omega' \end{vmatrix}$$

with like expressions for $CA' - C'A$ and $AB' - A'B$, and substituting, we have

$$\begin{vmatrix} P_a, & Q_a, & R_a, \\ A, & B, & C \\ A', & B', & C' \end{vmatrix} = w \begin{vmatrix} \cdot & x, & y, & z, & w \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & a_a, & a_a, & a_a, & -a_a \\ L, & M, & N, & \Omega \end{vmatrix},$$

or substituting for p_a, q_a, r_a their values $aw_a - ax_a, aw_a - aw_a, -aw_a + aw_a$, this is

$$\begin{vmatrix} \cdot & x, & y, & z, & w \\ A, & B, & C \\ A', & B', & C' \\ L, & M, & N, & \Omega \end{vmatrix}$$

whence, omitting the factor w , the equation is

$$\begin{vmatrix} p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta, & x, & y, & z, & w \\ a_a, & a_a, & a_a, & a_a, & -a_a \\ a_b, & a_b, & a_b, & a_b, & -a_b \\ L, & M, & N, & \Omega \\ L', & M', & N', & \Omega' \end{vmatrix} + 4p\alpha\beta \cdot p\beta\beta \cdot p\beta\beta \cdot p\beta\beta \cdot p\delta\beta \cdot p\delta\beta \begin{vmatrix} \cdot & x, & y, & z, & w \\ \cdot & a_a, & b_a, & c_a, & d_a, -e_a \\ \cdot & a_a, & b_a, & c_a, & d_a, -e_a \\ L, & M, & N, & \Omega \\ L', & M', & N', & \Omega' \end{vmatrix} = 0,$$

[Sadleirian $abc\mathcal{E}c$.]

C. VIII.

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where I recall that the $(L, \dots)$, $(L', \dots)$, $(L'', \dots)$, $(L''', \dots)$ are functions of (a, b) , (a, β) , (a, γ) , and (a, β) respectively. The values of the first two determinants thus are $p\delta\beta \cdot p\delta\beta$ and $p\delta\beta \cdot p\delta\beta$ respectively; that of the third is $= -p\alpha\beta \cdot p\delta\beta$, viz. this is $= -p\delta\beta \cdot p\delta\beta$, where, as before, $p\delta\beta = \dots$. And we have then the below-mentioned equation of the surface.

47. *Singularities.* The equation of the surface shows that (0) The point a is a 2-conical point; in fact, we have for this point $p\delta\beta = 0$, $p\delta\beta \cdot p\delta\beta = 0$; $p\delta\beta = 0$, $p\delta\beta = 0$. (1) The line a is a 4-tuple line: in fact, for any point on this line $p\delta\beta = 0$, $p\delta\beta = 0$; $p\delta\beta = 0$; $p\delta\beta = 0$, $p\delta\beta = 0$; whence the term in question, divides by $p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0$. (2) The line (a, β, γ) is a 4-tuple line: in fact, for any point on this line $p\delta\beta = 0$, $p\delta\beta \cdot p\delta\beta = 0$, $p\delta\beta = 0$, $p\delta\beta = 0$; whence the term in question, divides by w^2 . (3) The exorhospheric $aa\beta$, β is a simple curve: in fact, for any point of this curve we have $p\delta\beta = 0$, $p\delta\beta = 0$; whence the term in question intersecting in the lines a , β β and the curve $aa\beta\beta$. It is, moreover, obvious that the surface is touched along the curve by the hyperboloid $p\delta\beta$.

I notice that the surface meets the quartic $p\delta\beta$ in

lines (a, β, γ)	each 4 times,	12
„ (a, β, γ)	twice,	4
„ (a, β, γ)	, ,	4
curves $aa\beta\beta, \delta a$	, ,	8
		14 × 2 = 28

Surface octuple.

48. The equation of the surface may be written

$$p\alpha\beta - p\alpha\beta \cdot p\alpha\beta + p\alpha\beta - p\beta\beta \cdot p\alpha\beta \cdot p\beta\beta + p\delta\beta = 0;$$

where $p\delta\beta$ &c., is the equation of the quadric through the lines a , β , γ ; &c. $etpa$ has the form noticed above. The form is one of kk forms depending on the partitionment

$$\begin{pmatrix} a\beta \cdot \beta\delta \cdot \delta \\ \delta\delta \cdot \beta\delta \end{pmatrix}, \quad \begin{pmatrix} a\delta \cdot \beta\delta \cdot \delta a \\ \delta\beta \cdot \beta a \end{pmatrix}, \quad \begin{pmatrix} a\beta \cdot \beta\delta \cdot \delta a \\ \delta a \cdot \beta a \end{pmatrix}$$

of the six letters.

[*Sadleirian abc&c.*]

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Surface chiefly.

49. *Investigation.* The projections of the six lines are tangents to a conic: the condition for this is $(P, Q, R)\beta = 0$, where the last factor expresses, that the determinant obtained by writing successively (P_a, Q_a, R_a) , &c. for (P, Q, R) . The equation may be written

$$a\beta \cdot p\delta\beta - p\delta\beta - p\delta\beta - p\delta\beta - p\delta\beta - p\delta\beta + RZ \cdot p\delta\beta = 0,$$

where

$$a\beta = \begin{vmatrix} P_a & Q_a & R_a \\ P_b & Q_b & R_b \\ P_c & Q_c & R_c \end{vmatrix},$$

and substituting for $P, \dots$, &c. their values, we have $a\beta = a \cdot p\delta\beta$; whence the foregoing result.

50. *Singularities.* The equation shows that (1) The line a is a 2-tuple line: in fact, for the point of the line we have $p\delta\beta = 0, p\delta\beta = 0, p\delta\beta = 0, p\delta\beta = 0$. (2) The line (a, β, γ) is a 2-tuple line: in fact, for each point of the line we have $p\delta\beta = 0, p\delta\beta = 0$. (3) The quadripolyolic $a\beta\delta a + \beta b$ is a simple curve on the surface: for the $p\delta\beta = 0, p\delta\beta = 0, p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0$.

It may be remarked that the surface meets the hyperboloid $p\delta\beta$ in

lines (a, β, γ)	each twice,	6
„ (a, β, γ)	„	2
„ (a, β, γ)	„	2
curves $aa\beta\delta + \beta a$,		4
		$2 \times 8 = 16$

51. It might be thought that there should be on the surface some curve $a\beta\delta\beta$, such as the conics $a\beta\beta\delta$ on the surface octuply; but I cannot find that this is so. The equation of the surface is satisfied by two simultaneously (A being arbitrary)

$$p\delta\beta - p\delta\beta - p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0,$$

$$p\delta\beta - p\delta\beta - p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0;$$

which equations represent quartic surfaces, the first of these depending on the duo- line, and passing through the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ ($\eta + \beta + d = 0 - \beta = 0$ conditions, so that the equation of such a surface contains only an arbitrary parameter λ), and the second having β for a duo-line, and passing through the lines $\mathfrak{a}, \mathfrak{B}, \gamma$. But I am not able to determine so as to have these equations indicate a curve on the surface octuply; it is to be remarked that the line involving λ is, with $\beta\beta$ a curve on the free surfaces is, when intersect, but in only ten in 8 ($= 2 + 6$) conditions, of the line free passes through (3 conditions) and the lines (a, β, γ) each β times (5 conditions) intersection in lines (3 conditions). And as the surface is octuple, in $3 + 6 (= 18$ conditions) [*Sadleirian abc&c.*]

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The Exorhospheric $aa\beta\beta$, &c.

52. The notion is, that we have a fixed point a , two fixed lines $\mathfrak{a}, \mathfrak{a}$, and a single infinite series of lines, or any line of the generating lines of a slow surface; each generating line determines, with the point a , a plane; and if in this plane we draw, meeting the lines $\mathfrak{a}, \mathfrak{a}$, a line to meet the generating line in a point P , then the locus of this point P is the curve in question.

53. In the case in question, the singly infinite series of lines is that of the lines which meet each of the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ or say there are the generators of the hyperboloid $a\beta\gamma$ the lines, or any say *lc.*, $\mathfrak{a}, \mathfrak{B}, \beta$. as mentioned above on indeterminacy. It is not necessary for the purpose of the moment, but it is interesting to consider in conjunction therewith the exorhospheric arising in like manner from the intersections of the hyperboloid; it will appear that the two curves are the complete intersection of the quadric $a\beta$ by a particular surface. Observe that the two curves are given as follows: *viz.*, considering the the quadric $a\beta\beta$ as generated by either of the two systems of generating lines $\mathfrak{a}, \mathfrak{B}, \gamma$ on the lines $\mathfrak{a}, \mathfrak{B}, \gamma$, and on the other systems of the generating lines on the lines $a, \mathfrak{b}, \mathfrak{c}, \beta$ *viz.*, the lines (a, β, β) is a curve a in the surface β and (a, β, β) is a curve β in the surface β .

54. The points of either of the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ may be regarded as derived from the points of any of the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ by a $(1, 1)$ correspondence; *viz.*, the two generators of the same system of β which meet either of these lines β, β, β , the intersections with these lines are pairs of points, and the points on the two lines β, β , correspond to a univocal series; or write into this curve.

[Sadleirian $abc\&c.$]

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or, as the second equation may also be written,

$$(Q, R - QR_3)(P_a P_a a_a - P_a P_l a_a) - (P_a P_3 - P_l)(Q_a R_a - Q_l R_3) = 0;$$

viz. the second equation represents a cubic surface having upon it the lines ($P = 0, R = 0$) and ($Q = 0, P_3 = 0$); further, intersecting the quadric $PR - QR = 0$ in these lines, and besides in an exorhospheric curve, which is the required locus.

55. Representing the determinants

$$\begin{vmatrix} P, & Q, & R, & l \\ P_a, & Q_a, & R_a, & l_a \\ P_b, & Q_b, & R_b, & l_b \\ P_c, & Q_c, & R_c, & l_c \end{vmatrix}$$

by (a, b, c, d, e, f, g) &c., *viz.* values, we have, *e.g.*, $a = QR_a - Q_l R_b$, c., the equations of the conies in question are reducible to the form ($P = QR = 0$, and $R = 0$), and besides in the exorhospheric curve, which is the required locus.

56. Representing the determinants

$$\begin{vmatrix} P, & Q, & R \\ P_a, & Q_a, & R_a \\ P_b, & Q_b, & R_b \end{vmatrix}$$

viz., the second equation represents a cubic surface having upon it the lines ($P = 0, R_3 = 0$) and ($Q = 0, P_3 = 0$); further, intersecting the quadric $PR - QR = 0$ in these lines, and besides in an exorhospheric curve, which is the required locus.

56. But the elimination may be performed as follows, *viz.* multiplying by P_a, P_a , from the first two equations in θ , multiplying by P_a, P_a , from the second and adding, and so with Q_a, Q_a , &c. we obtain

$$\begin{aligned} (Q_a - RR_a) + bP_a + (P_a + lRR_a)\theta + (a + \beta\theta + \beta\theta) &= 0 \\ (Q_a R_a) + bP_a + (P_l RR_a)\theta + (a + \beta\theta + \beta\theta) &= 0, \\ (Q_a - RR_a) + bP_b + (P_b - lRR_b)\theta + (a + \beta\theta + \beta\theta) &= 0, \end{aligned}$$

We have here

$$\theta = \frac{-c + b\theta}{a + b\theta}, \frac{c + b\theta}{d - \beta\theta};$$

or, what is the same thing,

$$b\theta + (a - b)\theta = c.$$

Using this equation, written in the form $(a + b)\theta = c$ to transform the first or third of the four equations in θ , we obtain

$$-(aP + abQ + bR)\theta - aQ + bR + R'P,$$

and using the same equation, written in the form $(d - \theta)\theta = -e$, to transform the second or fourth equation, we obtain

$$(eP_b - \beta Q + RP_b + a)\theta = -\beta Q - \gamma R + dP_b,$$

[Sadleirian abc&c.]

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and hence, eliminating θ , we obtain

$$(bQ_a + R_a + aQ_a + R_a - bP_a)(cQ_a + R_a - Q_a) = (bP_a + aQ_a + bR_a + bR_a)(-aQ_a + dP_b)$$

which, on being of the second order in $(a, \dots)$ represents a quartic surface. The equation contains underwritten by the interchange of (P, Q, R) with (P_a, Q_a, R_a) that vanish, the opposite sextants is the surface; that the line $\mathfrak{a}, \mathfrak{B}, \gamma$ are not similarly for $p\delta\beta, p\delta\beta, p\delta\beta$; so that, omitting the constant factor $V = ie$, the value of the operation becomes

$$= \text{four}(eABP + bQ + CdR + DSe) + AP + EQ + CR + DS = \text{four}(eAB - BdQ - BdR)P + a(b - aB) + d(Ad - E$$

57. I obtain this same result also as follows. Consider a point (P_a, Q_a, R_a, S_a) on the quadric surface; $(P_a, Q_a, R_a, R_a) = 0$, the tangent plane at the point is

$$PR + QR_a + RQ_a + SP_a = 0,$$

and if this passes through the point a , then

$$PR + bQ_a + aR + dP_a = 0.$$

The line which in the tangent-plane meets the lines b, a is given, as before, by the equations

$$\begin{aligned} P_a R_a - Q_a R_a + R_a Q_a + R_a Q_a &= 0, \\ P_a R_a - Q_a R_a + R_a Q_a + R_a Q_a &= 0. \end{aligned}$$

Remembering the significations of $(a, \dots)$, the last three equations give

$$\begin{aligned} S_a : R_a : -Q_a : -P_a &= -aQ_a + dP_a + ab, \\ &: aP_a + R_b R_a + ac, \\ &: aP_a + R_b R_a + ad, \\ &: bQ_a + R_b R_a + cd; \end{aligned}$$

and substituting these values in $S_a P_a - Q_a R_a = 0$, we have the above equation of the quadric surface.

58. Or again, changing the notation, I take the equation of the quadric surface to be

$$(a, b, c, d, f, g, h, l, m, n)(x, y, z, w) = 0,$$

or

$$ax^2 + by^2 + cz^2 + dw^2 + 2fyz + 2gzx + 2hxy + 2lzw + 2mwy + 2nwx = 0.$$

A tangent-plane hereof is

$$\xi x + \eta y + \zeta z + aw = 0,$$

where ξ, η, ζ, a , are any quantities satisfying the relation

$$(A, B, C, D, F, G, H, L, M, N)(\xi, \eta, \zeta, a) = 0,$$

the capitals denoting the inverse coefficients. Supposing that the tangent-plane passes through a fixed point a , coordinates $(\alpha, \beta, \gamma, \delta)$, we have

$$a\xi + b\eta + c\zeta + da = 0,$$

[Sadleirian $abc\ell'c.$]

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and if the equations of the lines $\mathfrak{B}, \mathfrak{c}$ are as below: $(a = 0, y = 0)$ and $(z = 0, w = 0)$: then for the line in the tangent-plane meeting the lines $\mathfrak{B}, \mathfrak{c}$ we have

$$\xi z + \eta w = 0, \quad \xi z + a w = 0.$$

These last equations may be represented by

$$\xi : \eta :: -\xi : \eta :: -z, -w$$

and, substituting these values, we have

$$(A, \dots)(\xi, \eta, -z, -w)^a = 0;$$

that is

$$(A\xi^2 - 2H\xi\eta + By^2 - 2F\xi w - Lyw - 2NLw^2 + 2Myw + Dw^2) \\ + (xy - 2Gxw - 2Mxw + Cz^2)\eta = 0;$$

that is

$$(A\xi^2 - 2H\xi\eta - By^2)\xi w \\ + (ay - zw + 2Hw^2)\eta z = 0;$$

the line being connected with the point a by the equation

$$a\xi + b\eta + c\zeta + da = 0.$$

Whence, eliminating ξ, η , we have the question and answer, viz. that an integral $(A\xi\eta - 2Hy\eta - 2By^2 - Fy\eta - Gw\beta + Mu) - Bw^2z(zw - 8Lw + (cy + dw))$

$$-(\beta z + dw)(Bw^2 - 2Mu - Cz^2 - \text{etc. } (yz - 2Lzw + (cy + 2dw) - 0,$$

an equation in (x, y, z, w) which has previously obtained, and which can be without difficulty identified therewith.

The Circular Curves are Geodesics.

59. I proceed to show that the circular curves are geodesics; viz. that an integral of the geodesic equation is

$$(E, F, G)(\xi, \eta)^a = 0.$$

Starting from this equation, we have

$$2(B\xi\eta + Bw^2 + 2By^2)(Bw^2 + Bw^2 + Bw^2) + E(\xi\eta + Bu^2)(Fuu + Buu) + Bw^2 + Bw^2 = 0.$$

Now the equation, writing therein $\xi\eta = F\eta = zw$, gives $\xi\eta + Bw^2$, then expansion may be written

$$4F\eta(\xi\eta + Buu) = 0;$$

the value of b being therefore $-b - 2E\xi - z$. The result then divided thus becomes $2Eu(\xi\eta - Bw)(Fu + bw^2) + (F\xi\eta - Bw + B\xi\eta + Bw^2)y$

$$= \Sigma(\xi\eta + Zw)(F\xi\eta + bw^2),$$

that is

$$2(E(\xi\eta + Bw))(2Eu - Fuu) + F(\xi\eta + Zw + Fwu^2)bw\} \\ + (ay + bw)(Fu + bw^2) - (zw + \xi\eta + dw)(Pu + 2F\xi\eta + dw^2)$$

viz. taking $\beta E\xi = (\xi w^2 + Fwu + d^2)$. This appears with the geodesic equation.

The foregoing integral, $(E, F, G)(\xi, \eta)^a = 0$, I believe, a particular, not a singular, solution of the geodesic equation.

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where Σ denotes the sum of the three terms obtained by cyclic interchange of x, y, z ; and we similarly have

$$\begin{aligned} p\beta\beta &= (z_1 - h\eta_1 z_1)(x - h\eta_1 x_1) - z_1 h\eta(z - bx), \\ p\beta\delta &= (z_1 - h\eta_1 x_1) - z_1(x - h\eta_1 x_1) - z_1 b\eta(z - bx), \end{aligned}$$

and similarly for $p\delta\beta$, see *post*, &c. 60. To obtain the intersection with $\delta\beta = w = 0$, writing $w = \frac{1}{2}\theta$, then

$$\begin{aligned} p\alpha\beta &= \{a_a - hc_a - \tfrac{1}{2}\theta(z_a - hya)\} b(x - h\delta z), \\ p\delta\beta &= \{a_a - he_a - \tfrac{1}{2}\theta(z_a - hya)\} b(x - h\delta z), \end{aligned}$$

or say

$$\sqrt{p\alpha\beta \cdot p\delta\beta} = M(z - h\delta\eta).$$

also the expression in [] become

$$\begin{aligned} M_a &= (a_a - hc_a) + \tfrac{1}{2}\theta(z_a - hya)\} b(x - h\delta z), \\ &\quad - (a_a - hc_a)(z_a - hya) - h(z - bx)(a_a - hc_a) \end{aligned}$$

so that the norm in question is

$$\text{Norm } \sqrt{M_a(x - h\delta\eta) - h(x - bx)(a_a - hc_a)\} b - hgM_a,$$

where M_a is now considered to stand for

$$\{(a_a - hc_a) + h\}(a_a - hc_a) + h(a_a - hc_a) b(x - h\delta z) b - h(2g)(a_a + a_a) - 2bx,$$

where M_a is now considered to stand for

$$\{(a_a - hc_a) + h\}(a_a - hc_a) + h(a_a - hc_a) b(x - h\delta z) b - h(2g)(a_a + a_a) - 2bx,$$

Observing that the norm was originally the product of 8 factors, this breaks up into:

$$(b\eta + (a - n)\beta + bw^2)(a_a)\beta(x - h\delta\eta_1) b - h\delta z = 0,$$

where the new term is the product of the four factors. 61. Writing for greater convenience a, b, c, a in place of h_a, h_b, h_c, h_d , and observing that M_a is a quadric function of h_a, h_b , that is of h , the last-mentioned norm hsh which is the surface, then takes the form: that the two above each touch the quadric surface, or that the lines are tabulated.

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[Continuing with]

$$A = PP', \quad B = (PQ' + P'Q), \quad C = QQ';$$

then

$$4AC = (PQ' - P'Q)^2,$$

and we have

$$\begin{aligned} P, \quad Q &= a_a w x_a, \quad b_a y_a, \\ P, \quad Q &= a w x_a, \quad b y_a; \end{aligned}$$

whence

$$\begin{aligned} PQ - PQ' &= (xyx_a - x_a y_a)w \\ &= \{xyx_a - xyy\} \{xyw_a x w_a w x_a\} \\ &= (xyy - xyx)w. \end{aligned}$$

viz. if (a, b, c, f, g, h) are the coordinates of the line ab , this is

$$= bs a + bx bs aw.$$

Hence, omitting the constant factor $(yz - x)^2 + (zx - y)^2$ (that is $(b_3 - b_3, h_3) - (g_3 - g_3, h_3, h_3)$), the foregoing equation must = 0 becomes

$$(bs a - g bax)^2 = a b b b b c ba (w x_a y w_a - xy a y_a),$$

the equation in question of the quadric with the surface are obtained by combining the equation $w x_a - g w = 0$ with the several equations

$$(bs a - g f bax)^a + a bs (xy w_a - xy a x_a) = 0,$$

$$(bs a + g bax)^a + a bs (xy w_a - xy a x_a) = 0,$$

viz. these are

lines $(a, \beta, \gamma, \delta)$ each 8 times 16

lines $(a, \beta, \gamma, \delta)$ each 2 24

lines $(a, \beta, \gamma, \delta)$ each 2 24

lines $(a, \beta, \gamma, \delta)$ 8

$$(16 + 24 + 8) \times 2 = 48 \quad \times 8$$

But it is clear that the lines $(a = 0, w = 0)$ and $(a = 0, z = 0)$ are introduced by the process of elimination, and are no part of the intersection. The complete intersection consists of the lines $(a, \beta, \gamma, \delta)$ each β times, the lines (a, β, γ) each β times, and the lines $(\beta, a, \beta, \gamma)$ each β times. The above last-mentioned line being only doubly lines on the surface, this means that the two sheets each touch the quadric surface, or that the lines are tabulated. C. VIII. 18

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On the Mechanical Description of Certain Sextic Curves.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 105–111.
Read April 13, 1872.]*

THE curves in question might be taken to be those described by a point C rigidly connected with points A and B , each of which describes a circle; but the construction is considered under a somewhat more general form. I consider a quadri-lateral, the sides of which are a, b, c, d , and the inclinations of these to a fixed line $\alpha, \beta, \gamma, \delta$. This being so, if a, b, c, d are one of the angles, say δ , are constant, then we have between the three variable angles the relation

$$a \cos \alpha + b \cos \beta + c \cos \gamma + d \cos \delta = 0,$$

$$a \sin \alpha + b \sin \beta + c \sin \gamma + d \sin \delta = 0;$$

giving rise to a single relation between any two of the variable angles; and we consider a curve such that the coordinates x, y of any point thereof are given linear functions of the sines and cosines of the three variable angles, or, what is the same thing, of the sines and cosines of any two of these angles. We then unite together what would otherwise be distinct cases; for everything is symmetrical in regard to the sides a, b, c and the corresponding variable angles α, β, γ , irrespectively of the order of succession of these sides; and we can then in like manner connect together, in any way two or pleasure, say α, β of the variable angles, without determining whether the sides a, b are conjugate or separate.

Eliminating then, the variable angle γ , we obtain between α, β a relation which, if we write therein $\tan \frac{1}{2}\alpha = u, \tan \frac{1}{2}\beta = v$, takes the form $(u^2 + v^2, V = 0$; the value of the variables α, β is expressible rationally in terms of the other of them and of the root of a quartic function thereof; say α is a rational function of v and $\sqrt{V}$. And

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hence a curve for which the coordinates x, y are rational functions of u, v is a curve having a deficiency $D = 1$, or, what is the same thing, having a number of dps. less by unity than the maximum number $[= \frac{1}{2}(n-1)(n-2)]$, if n be the order of the curve].

It will further appear that the relation $(\ast \tilde{a} u, 1)^2(\ast, 1)^2 = 0$ is satisfied by the values $u = v = i$ and $u = v = -i$ (if, as usual, $i = \sqrt{-1}$).

In the curves in question, the coordinates (x, y) are given linear functions of the sines and cosines of α, β ; and if we make the curve meet an arbitrary line $Ax + By + C = 0$, we obtain between the sines and cosines of α, β a linear relation which, substituting therein the expressions in terms of u, v , takes the form

$$(\ast \tilde{a} u, 1)^2(\ast, 1)^2(1 + v^2) + (\ast, 1)^2(1 + u^2) = 0,$$

viz. this is a relation of the form $(\ast \tilde{a} u, 1)^2(\ast, 1)^2 = 0$, such that it is satisfied by the four sets of values $u = \pm i, v = \pm i$, and therefore in particular by the values $u = v = i$ and $u = v = -i$.

Hence, considering the intersections of the curve by the arbitrary line, the values of (u, v) are given by the two equations $(\ast \tilde{a} u, 1)^2(\ast, 1)^2 = 0, \tilde{a}(\ast u, v), V\tilde{a}(\ast v, 1) = 0$, regarding for a moment u, v as ordinary rectangular coordinates, represent each of them a quartic curve having two dps. at infinity on the axis $u = 0, v = 0$ respectively; each of these points reckons therefore as 4 intersections, and the number of the remaining intersections therefore is $4.4 - 2.4 = 8$. But, by what precedes, the two quartic curves have also in common the points $u = v = i$ and $u = v = -i$; and rejecting these, there remain $8 - 2 = 6$ intersections.

The conclusion is, that the curve is a sextic curve of deficiency 1, that is, having 9 dps. The reasoning may be presented under a slightly different form as follows; regarding u, v as coordinates, we have the curve $(\ast \tilde{a} u, 1)^2(\ast, 1)^2 = 0$, a bicodal quartic curve, and having therefore the deficiency 1; the curve passes, as above-mentioned, through the points $u = v = i$ and $u = v = -i$. The required curve is obtained as a transformation of the quartic curve by formulæ of the form $x : y : z (= 1) = P : Q : R$, where P, Q , and $R [= (1+v^2)(1+u^2)]$ are quartic functions of the coordinates u, v , such that $P = 0, Q = 0, R = 0$ are each of them a quartic curve passing twice through each of the nodes and once through each of the before-mentioned points, $(u = v = i)$ and $(u = v = -i)$. The factor thus is

$$I_n = pcd. - 1\Sigma p\Sigma a b a.$$

$p\beta\beta = p^a, a+p^b, b+p^c, c+p^d, d, p\beta = 0$ is a binodal quartic curve, and the maximum number of dps. of and the entire defect of the cubic in u, v is one, the deficiency of $u, v = 1$.

To obtain the foregoing equation $(\ast \tilde{a} u, 1)^2(\ast, 1)^2 = 0$, the elimination of γ gives:

$$(a \cos a + b \cos \beta + d \cos \delta + (c \sin a + b \sin \beta + d \sin \delta) = c^2,$$

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that is

$$a^2 + b^2 + d^2 + 2ab \cos(a - \beta) + 2ad \cos(a - \delta) + 2bd \cos(\beta - \delta) = c^2,$$

or, substituting herein the values

$$\cos a = \frac{1 - u^2}{1 + u^2}, \quad \sin a = \frac{2u}{1 + u^2}, \quad \cos \beta = \frac{1 - v^2}{1 + v^2}, \quad \sin \beta = \frac{2v}{1 + v^2},$$

this is easily found to be

	1	2v	v ²
1	$(a + b)^2 - 2(a + b)d \cos \delta + a^2$	$2bd \sin \delta$	$(a - b)^2 - 2(a - b)d \cos \delta + d^2 - c^2$
2u	$2bd \sin \delta$	$4ab$	$2bd \sin \delta$
u ²	$(a - b)^2 - 2(a - b)d \cos \delta + d^2 - c^2$	$2bd \sin \delta$	$(a + b)^2 + 2(a + b)d \cos \delta + d^2 - c^2$

or setting, for greater convenience, $\delta = 0$, that is, taking a, β, γ to be the inclinations of the sides a, b, c to the fixed side d , this is = 0.

	1	2v	v ²
1	$(a + b - d)^2 - c^2$	0	$(a - b)^2 - d^2 - c^2$
2u	0	$4ab$	0
u ²	$(a - b)^2 - d^2 - c^2$	0	$(a + b)^2 + d^2 - c^2$

or say this is

$$(A + Bu^2) + 4abuv + v^2(C + Du^2) = 0,$$

where, writing for shortness

$$\begin{aligned} a + b - c - d &= \lambda, & -a + b + c + d &= \lambda', & -a - b + c + d &= \mu', \\ a + b - c + d &= \lambda', & +a + b - c + d &= \mu', & +a + b + c + d &= \mu', \\ a + b + c + d &= \nu, & -a + b + c + d &= \nu', & +a + b + c + d &= \nu', \end{aligned}$$

then

$$A = -\lambda\nu, \quad B = -\lambda\mu', \quad C = \nu\lambda', \quad D = \mu\nu'.$$

We at once verify that the equation is satisfied by $u = v = \pm i$, viz. this will be so if only

$$A - B + C - D = 0;$$

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and we have

$$\begin{aligned} A + B &= 2(a + b)^2 - 2c^2 - 2d^2, \\ B + C &= 2(a - b)^2 - 2c^2 - 2d^2; \end{aligned}$$

whence the relation in question.

Writing $u = v$, we have

$$\begin{aligned} v^2(C - B) + 4abv^2 + Cv^2 &= 16(b^2 + b^2)(a^2 - B^2) - (s^2 - C); \\ &= -j(A + 4)D + B - C, \end{aligned}$$

whence, that, corresponding to $u = v$, we have the values $u = v = \pm i$ and $u = v = \pm \frac{A-B}{C-D}$, and corresponding to $u = -v$, the values $v = -i$ and $u = -v = \pm \frac{A+B}{C+D}$.

And in like manner, corresponding to $v = i$, we have the values $u = i$ and $u = v = \pm \frac{A-C}{B-D}$, and to $v = -i$, the values $u = -i$ and $u = -v = \pm \frac{A+C}{B+D}$.

It is easy to show that, if $u = v = a$, $r = v$ are then values connective to $u = v = i$, then $u = b(2 - 1)$ in fact, substituting the foregoing values in the relation between u , v , and writing for that the value, we have

$$A(b - a) + B(a + 1) + a(d + 1) + b(b - 1) + Bb(a + 1) + Cb(b - 1) + Ds(b + 1) = 0,$$

which is

$$= s(A + Bs - C - Ds) + s(A + B + Cs - D) = 0.$$

or finally $a = -2v$, if $u + v = 0$. And similarly, if $a = -i$, then $a = i = -v$, viz. $u + v = 0$; then we have shown that the points which we have found in the surface are connected with the surface, and the form of pairs $u = -v$.

The points in infinity on the sextic curve are those for which $1 + a^2 = 0$ or $1 + b^2 = 0$, or each of these, is 0; viz. the values of u, v for the six points are as follows: 1° $a, u = +1, v = -i, +1, -i, +1, -i, +1$;

$$u = -\frac{ai + a}{c - a}, \quad v = +\frac{a + ic}{c - a}, \text{ four pair of points; } 2^\circ v = i (= i), u = -i (= i), \text{ two pair of points; } u = +\frac{a - i}{c - a}, \quad v = -\frac{a + ic}{c - a}$$

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then, if $u = -i(v = i)$, then $u = -i$, $v = i$ are pair of points, viz. δ values of u , v . Hence the points on the sextic curve at infinity are as follows:

$$\Sigma^\circ. u = \frac{1}{4}(a + i) \text{ pair of points;}$$

$$\Sigma^\circ. v = \frac{1}{4}(b + i), \text{ pair of points;}$$

$$\Sigma^\circ. \text{etc.}$$

which six points in question denote three pair of points, viz. the circular points at infinity.

The foregoing values of A, B might be said to be “circular grand u ” if $a = c$, $a = -c$; and similarly to be “circular grand B ” if $b = d$, $b = -d$.

And we see at once that if the curve are circular grand u , then the four pair of points coincide with the circular points at infinity; and that, in like manner, if the curve are circular grand B , the second pair of points coincide with the circular points at infinity. If the curve are circular grand u , the third pair of points coincide with the circular points at infinity. But, if the curves are circular grand B and u , u also, then is a unicursal sextic having on each of the four sides u , b , c d a node: the condition in fact is

$$AD + BC - 16ab^2 = 0;$$

that is

$$\lambda\lambda' + \lambda'\lambda - \lambda'\lambda + 4\sqrt{ABCD} = 0,$$

where, putting for a moment $M = a^2 - b^2 - c^2 + d^2$, we have

$$\lambda\lambda' + \mu\nu' + \nu\lambda' = (M - 2ab)^2 + (M + 2ab)^2 + (M + 2ab)^2 - 2ab(M + 2ab)^2 = 3M^2 - 4(a^2 + b^2)d^2,$$

and thence

$$\lambda\lambda' + \nu\lambda' - \lambda'\lambda = -a^2 - b^2 + a^2 + b^2 + a^2c^2 + bd + 4cdM,$$

$$-p\nu p' = \nu^2 - 4a^2b^2 - \nu^2 + 4a^2b^2 + 4a^2 + bd + 4cdM,$$

viz. the equation is

$$4cdM = -a^2 - b^2 - c^2d^2 + b^2cd - 2cd(a^2b^2c^2 - 2cdM).$$

the deficiency, -1 , being then $= 0$.

Hence the equation simultaneously $1^\circ. a = c, b = d$; $2^\circ. b = d, a = c$; $3^\circ. a + d, a + b$; these cases are easily equivalent, but on reading, give a third such, which may be a remarks, case, or arrival. In the third-mentioned case, the relation between a, b contains the factor a , and throwing this out, and also the constant factor $4a$, it is

$$4(a + a^2)(a + 1)(a + 1) + (a + 1)(a + 1)(a + 1) + 2aa = 0,$$

viz. a is given as a rational function of b .

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It would at first sight appear that the curve might become unicursal in a different manner; viz. it would be unicursal if

$$1l\lambda\lambda' = (A + Bv)^2(C + Dv)^2,$$

was a perfect square; but this is only the case when one of the four sides a, b, c, d is $= 0$. The condition in fact is

$$AD + BC - 16ab^2 = \pm 2\sqrt{ABCD};$$

that is

$$\lambda\lambda' - \mu\nu' + \mu'\lambda' - \lambda'\lambda + 4\sqrt{ABCD} = \mp 4\sqrt{ABCD},$$

where, putting for a moment $M = a^2 - b^2 - c^2 + d^2$, we have

$$\lambda\lambda' - \mu\nu' + \mu'\lambda' - \lambda'\lambda = 4ab^2 - 4bacd + 4abd + 4cdM,$$

and thence

$$\lambda\lambda' - \mu\nu' + \mu'\lambda' - \lambda'\lambda = -4cd - abcd(cd - ab) + 8cdM;$$

viz. putting for a moment $X = M^2 - 4cd(cd + 2b)$, this is

$$(X - 4cdM)^2 = (X + 4cdM - 16abc^2d)^2;$$

viz. substituting for X its value, the equation is

$$16cd(a^2 + b^2 - c^2 - d^2)(M + 2cd) = 0;$$

that is

$$16abcd(M + 2cd) = 0,$$

which means simultaneously $1^\circ. a = 0, b = c; 2^\circ. b = c, a = c; 3^\circ. c = 0, a = c$; the three cases are easily equivalent, but the resulting general theorem is in different forms. 11. Here $A = 0, B = 4ac(c + b), C = 0, D = 4ac(c + b)$; the relation between a, b contains the factor a , and throwing this out, and also the constant factor $4a$, it is

$$u(v + b) + 1(a + b + 1) + 2a = 0,$$

viz. a is given as a rational function of b .

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2°. Here $A = 0, B = 4(a + c), C = 4b(a + c), D = 4b(a + c)$; the equation contains the factor a , and throwing out this and also the constant factor $4b$, the equation is

$$v(b - a) - a(a + b + 1) + 2a + a = 0;$$

viz. v is then given as a rational function of a . 3°. Here $A = 4(a - b), B = 0, C = 4b(a - b), D = 0$; or, dividing by $4a$, the equation is

$$(a - b)(1 + v^2) + (a + a + b)(1 + b^2) = 0,$$

which may be reduced to

$$(a + 1)(a + b) + av + a = 0;$$

giving a or v each a rational function of the other. I do not discuss the theory in detail, but only remark that, in each case there is a conic thrown off, and that in place of the sextic we have a unicursal (or trinodal) quartic curve.

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On the Surfaces divisible into Squares by their Curves of Curvature.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 120, 121.
Read June 13, 1872.]*

PROFESSOR Cayley gave an account of an investigation recently communicated by him to the Academy of Sciences at Paris. The fundamental theorem is that, if the coordinates x, y, z of a point on a surface are expressed as functions of two parameters p, q (each expression, of course replacing the equation of the surface), and if these parameters are such that $g = \text{const.}, q = \text{const.}$ are the equations of the two sets of curves of curvature respectively; then (writing for shortness

$$\frac{dx}{dp} = a_1, \quad \frac{dx}{dq} = b_1, \quad \frac{dy}{dp} = a_2, \quad \frac{dy}{dq} = b_2, \quad \frac{dz}{dp} = a_3, \quad \frac{dz}{dq} = b_3;$$

A, B, C for $bc - cb, ca - ac, ab - ba$).

In order that x, y, z , the coordinates x, y, z considered always as functions of p, q , satisfy the equations

$$\begin{vmatrix} a_1 & b_1 & A_1a_1 + Bb_1 \\ a_2 & b_2 & A_2b_2 + Bb_2 \\ a_3 & b_3 & A_3b_3 + Bb_3 \end{vmatrix} = 0.$$

The last equation is equivalent to

$$a_1 + A_1c_1 + Bb_1 = 0,$$

$$a_2 + A_2b_2 + Bb_2 = 0,$$

$$a_3 + A_3b_3 + Bb_3 = 0,$$

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and if in the notation of Gauss we write

$$a_1^2 + a_2^2 + a_3^2 = E,$$

$$a_1^2 + a_2^2 + a_3^2 = G,$$

then adding the equations multiplied by a_1, b_1, c_1 respectively, and also adding the equations multiplied by a_2, b_2, c_2 respectively, we find

$$A = \frac{1}{E} \frac{dE}{dq}, \quad B = -\frac{1}{G} \frac{dG}{dp};$$

and the equations thus become

$$2a_1 - \frac{1}{E} \frac{dE}{dq} a_1 + \frac{1}{G} \frac{dG}{dp} b_1 = 0,$$

$$\&c., \quad \&c., \quad \&c.,$$

which, in fact, agree with the equations (10 bis) in Lamé's "Leçons sur les coordonnées curvilignes," Paris (1859), p. 93. The motion will be divisible into squares if only $E : G$ is the quotient of a function of p by a function of q , or say if

$$E = \Phi p^2, \quad G = \Psi q^2,$$

where Φ is any function of p, q (any of p and q respectively). It seems likely indeed that $E : G$ is the quotient of a function of p by a function of q , expressing therefore the above equations may be reduced to

$$\frac{1}{E} \frac{dE}{dp} - \frac{1}{G} \frac{dG}{dq} a_1 + \frac{1}{E} \frac{dE}{dq} b_1 = 0,$$

and the equations for x, y, z are

$$2a_1 - \frac{1}{dp} \frac{d}{dp} a_1 + \frac{1}{dq} \frac{d}{dq} b_1 = 0,$$

viz. x, y, z being homogeneous functions of p, q such that $x = a_1 q + b p$, and which besides satisfy these equations, or say which each of them satisfies the equations

$$2a_1 - \frac{1}{dp} \frac{d}{dp} a_1 + \frac{1}{dq} \frac{d}{dq} b_1 = 0,$$

then the values of x, y, z in terms of (p, q) determine a surface which has the property in question.

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On the Mechanical Description of a Cubic Curve.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 175–178.
Read November 14, 1872.]*

IF the coordinates x, y of a point on a curve are rational functions of $\sin \phi, \cos \phi, \sqrt{1 - F \sin^2 \phi}$, the curve has the deficiency 1, and conversely in any curve of deficiency 1 the coordinates x, y can be thus expressed in terms of the parameter ϕ . Hence writing $\sin \phi = b \sin \theta$, the coordinates will be rational functions of $\sin \phi, \cos \phi$, or say of $\sin \phi, \cos \phi, \sin \theta, \cos \theta$; and the mechanical representation of the relation $b \sin \phi = \sin \theta$ is, writing for shortness $I = ar + b$, that of any curve $y = a + b \sin \phi$, which values may be written

$$x = a \sin \phi,$$

$$y = \frac{1}{\sin \phi} (\cos \theta - 1) = \sin \theta;$$

I found, however, that this was not the cubic curve most easily constructed; and I ultimately devised a mechanical arrangement consisting of: 1. Rod GH , and connected with it by a pin at H , rod HI (*).

(*) *There were a mechanical conventions in this, but obtuse that producing GH to meet IF in I , the angle GIF might have been made one of.*

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2. Square ACD , and connected with it by a pin at D , rod DG .
3. Square ECF , the two squares being connected by a pin at C .
4. Rod IJ .

[Figure: Mechanical linkage with quadrilateral, two squares, and rods — diagram omitted]

The rod GH rotates about a pin at G ; taking $HA = HJ$, there is a pin at A connecting a fixed point of this rod with the extremity A of the square ACD : the fixed point B of the square moves along the line Gx . There is a pin at J connecting the extension of the rod HI with the rod JL , the angle which it makes with JL being equal to that which the rod IJ makes with JG , the fixed point F of the square ACF also slips along the line Gx , and the two squares are so related to each other that, in the motion of the system, while the curve traced out by the point L rotates about a fixed point, the latter of the square $ABCD$ also rotates by reason of its equation of the form above referred to; and then the lines DJ , CF keeping the connection with each other determined by the relation between the squares and the pin at I , the rod IJ rotates about J . The angles which the rods make of being all 8 to it.

I write

$$\begin{aligned}\angle DBJ &= \theta, & \angle ABJ &= \phi; \\ GA &= a, & AB &= b, & AC &= c, & CD &= d, \\ AB &= HJ = \mu.\end{aligned}$$

Take

$$x = QC, \quad y = JL.$$

I write

$$\angle DBJ = \theta, \quad \angle ABJ = \phi,$$

$GA = a$, $AB = b$, $AC = c$, $CD = d$, $AB = HJ = \mu$. Take

$$x = QC, \quad y = JL.$$

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We then have $x \sin \theta = b \sin \phi$, and moreover the length AJ being $= b \cos \phi \pm b \cos \phi$, and thence $HJ = b \cos \phi$, $HJ = b \cos \phi$, we have

$$y = (b \pm a) \frac{-b \cos(\theta \pm \phi)}{\sin \theta} + b \cos \phi,$$

$$x = QC = c \sin \phi;$$

whence also

$$xy = b - b \sin(\theta \pm \phi),$$

or we have

$$xy = b - b \left(1 - \frac{x^2}{c^2}\right) \left(1 - \frac{b^2}{d^2}\right) \pm b \cdot \frac{x}{d} \frac{x}{c};$$

that is

$$x \left(y - \frac{bd}{cd}\right) = b - b \left(1 - \frac{x^2}{c^2}\right) \left(1 - \frac{b^2}{d^2}\right),$$

or rationalising and reducing, this is

$$xy^2 \frac{bdd}{cd} 2bcy + \left[b \left(1 - \frac{b^2}{c^2}\right) - 2b \right] c^2 + a^2 - b^2 = 0,$$

a quartic curve with two dps.

In the particular case $F = a = -b$, the equation becomes

$$\left(y - \frac{2b}{c}\right) - a + b(xy - d(a+b)x) = 0,$$

the equation in x becomes the equation in the form

$$x \left(y - \frac{2b}{a}\right) d(a-b) [x^2(a-d) + abk] x;$$

the second factor is extraneous, and the curve is thus the cubic

$$x \left(y - \frac{2b}{a}\right) d(a-b) [x^2(a-d) + abk] x;$$

at one stage appears from the foregoing irrational form of the equation.

In the particular case $b = c$, the equation contains the factor x , and omitting this it becomes

$$x \left(y - \frac{2bc}{a}\right) cdy + bcb + \dots$$

viz. we have a cubic curve with three real asymptotes meeting in a point which is also the centre of the curve.

If simultaneously $a = b$ and $b = c$, then equation becomes

$$x \left(y - \frac{2b}{a}\right) cdy - 2cdx = 0,$$

the actual locus being in this case the line $y - \frac{2bd}{c} = 0$.

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Writing $b = c$, and for greater convenience $b = c = a = -d = 1$; also for the sines supposing $b < c$, and writing $a = 1, \sin b$, then we have

$$\sin \phi = h \sin \theta, \quad \sin \phi = a \sin \phi,$$

that is

$$\sin \phi = h \sin \phi,$$

and

$$x = c \sin \phi, \quad y = \frac{1 - \cos(\theta + \phi)}{\sin \theta} - \cos \phi,$$

that is

$$xy = 1 - \cos(\theta + \phi) - c \sin \theta \cos \phi,$$

giving the rationalized equation

$$x(y - 2(1 - c)x - 2y) + 4x^2;$$

the angle ϕ may be anything whatever, but θ varies between the limits $\pm \frac{1}{2}\lambda$, the simultaneous values of these angles and of the coordinates being

$\phi = 0$	$\theta = 0$	$x = 0$	$y = 0$
$\phi = 90^\circ$	$\theta = \lambda$	$x = c$	$y = 1 + \sin \lambda$
$\phi = 180^\circ$	$\theta = \pi$	$x = 0$	$y = 2$
$\phi = 270^\circ$	$\theta = -\lambda$	$x = -c$	$y = -(1 + \sin \lambda)$
$\phi = 360^\circ$	$\theta = 0$	$x = 0$	$y = 0;$

and it thus appears that the mechanism gives the continuous branch which belongs to the asymptote $x = 0$ of the cubic curve; the other two branches belong to $x = +\sin \phi$, $y = \frac{1 - \cos(\theta + \phi)}{\sin \theta} - \cos \phi$, which would require a slight alteration in the arrangement of the mechanism.

I remark that if AB, HI had been unequal, then writing $\angle HJA = \chi$, this would be connected with θ and ϕ by an equation of the form

$$\sin(\theta + \phi) = m \sin \chi,$$

and the coordinates x, y would be rational functions of the sines and cosines of θ, ϕ, χ ; the deficiency is in this case = 3.

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On the Mechanical Description of Certain Quartic Curves by a Modified Oval Chuck.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 186–190.
Read December 12, 1872.]*

THE geometrical principle of the oval chuck is the well-known one that if a plane move in such manner that two lines Ox, Oy , fixed in the plane and meeting with it, pass through two fixed points A, B respectively; then any fixed point P traces out on the plane an ellipse. The point A is on the (geometrical) axis of the mandrel; there is connected with the head a guide-ring moving horizontally; the point B is the centre of the guide-ring; the head is fixed in a ring through a ring which slides horizontally at right angles to the axis of the mandrel; and as the axis revolves, the centre B describes practically a circle on the head A , which by reason of the guide-ring remains fixed at the centre. Hence the point P in the plane traces out the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, as it ought; but as the axis of A is a fixed point, this point being the line at infinity in the plane, what would be for a lathe.

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The apparatus consists of a piece of rock-board, about 10 inches long by 7 inches broad, pierced with a circular hole of 1 inch diameter for a vertical axis: the edges of the board serve as guides for the frame L , which carries the guide-ring; and resting on the board we have the frame M itself guided by the frame L so as to be able to move independently of each other, but they can be clamped together; the axis has

*[Figure: Modified oval chuck with rectangular board, frames, axis, slide rings, and slot —
diagram omitted]*

upon it a square and, the sides of which work on the slot of an eccentric, the frames of this being adjustable by means of a screw passing through the nut and arm; and there is above the eccentric a square nut shown in the figure. This is capable of rotation round an axis at the foot, of an arm at the action by means of a slide; the arm is checked at the back of the eccentric, the action being placed so that the eccentric may be placed in two different positions: viz., the guides may be either upper or lower edges of the cut, those which are parallel to the sides of the slot, or the front of the slot are at right angles to it.

Supposing the bed planned as above upon the guide-ring and nut, then if the frames L and M are disconnected, and the former of them is fixed, the frame M will, on rotation of the axis, be carried backwards and forwards by the encore; but this will in no wise affect the motion of the head, but the arrangement is then equivalent to the oval chuck, and a fixed point in the head will go gives a position will trace out the frame M alone, then we have always through the point A , B respectively; the former of these being a fixed point, the latter of them a point moving according to determinate law backwards and forwards along a fixed line through A .

The plan is carried out in a drawing apparatus which I have had constructed in wood, the axis being here vertical instead of horizontal, and the details of course different from what they would be for a lathe.

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Let the coordinates of the fixed point P , referred to axes through A , the first of them perpendicular to, and the second coincident with, AB , be h, v ; let the distance AB be $= a$, and let θ denote the angle BAO . Then, if x, y are the coordinates of P referred to the origin O and axes Ox, Oy , we have

$$\begin{aligned}x &= a \cos \theta + h \sin \theta + v \cos \theta, \\y &= -h \cos \theta + v \sin \theta,\end{aligned}$$

which, if a be constant, gives a quadric equation, or the curve is an ellipse; and, in particular, if $h = 0$, that is if the point P is on the line AB , then we have

$$x = a(c - r) \cos \theta, \quad y = v \sin \theta,$$

or the curve is

$$\frac{x^2}{(a + r)^2} + \frac{y^2}{r^2} = 1.$$

But if a is a given function of θ , then this equation is still found by eliminating θ between the two equations for x and y . In particular, if the distance AB is equal to the projection upon the tangent of a circle, as shown, in the figure, then if b be the radius AC of this circle, and ϕ the inclination of AC to $A-O$ (it being taken to be constant), we have

$$a = b \cos(\theta + \lambda),$$

and the equation are

$$\begin{aligned}x &= b \cos(\theta + \lambda) - b(1 \cos(\theta + \lambda)) + v \cos \theta, \\y &= -b \cos \theta + v \sin \theta.\end{aligned}$$

The elimination is easily done in as it were $= a + 0$; viz., we may determine y, z in such wise that

$$\begin{aligned}b \sin \theta + v \cos \theta &= u(\theta + \lambda) - y \cos \theta, & -y \cos \phi &= \sin \theta, \\-b \cos \theta + v \sin \theta &= a(\theta + \lambda) - y \cos \phi &= -y \sin \theta;\end{aligned}$$

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and then

$$x = y \cos \theta - b(1 - \cos \theta + a) + a(\theta + a) - y \cos \phi, \quad y = -y \sin \theta - x;$$

1, will, for greater simplicity, at once write $b = 0$; the equations than are

$$x = -y \cos \theta + a(\theta + a) \sin \theta \cos \theta, \quad y = y \sin \theta;$$

or say there is

$$x = (a + b \sin \phi) \cos \theta + b \sin \theta \cos \theta,$$

$$y = y \sin \theta,$$

whence, we have

$$x \cos \phi = y \sin \phi + b(1 - \sin \theta - F \cos \theta \cos \theta + b \cos \theta b)$$

that is

$$(x \cos \phi - y \sin \phi - \frac{1}{2}b)^2 = b^2 by \sin \theta V (F - b);$$

an equation which will assume a more simple form if either $\lambda = 0$ or in $\lambda = 90^\circ$; that is, if in the apparatus the small-circle which guides the bed are either at right angles or parallel, to the slot of the slot.

Taking the general case, and writing for convenience $\cos \phi = \text{const.} \cos \phi = a$, the form is given by the form

$$x = e(\eta, b, B, Y)$$

$$y = \eta = b \cdot a, Y$$

$$F = a + b \cdot a Y;$$

viz. the elimination of $\xi \eta$ from these equations leads to the equation of the curve. The points of the curve have then $u \sin \phi, (\eta + a)$ correspondences with those of the conic $F = \eta = b \cdot a$, viz. the circle being unicursal, the curve is also unicursal. And considering the relation between the variable parameters ξ, η as the equation of a conic in the ξ, η plane, the curve is in like manner the unicursal transformation by the conic ξ, η so that the foregoing relation form a unicursal quartic. The points where the line at infinity in the ξ, η plane meet the conic conic correspond to the points ξ, η at infinity ($a = 0$), ξ, η at infinity, then $\xi =$ to be a position $\xi = \xi + Z\xi$, viz., $\xi + \xi$ at the point corresponding to the intersection of $\xi + Z\xi = \xi + \xi$ by the quadric $\xi = \xi$ or ξ , that is, $\xi + \xi, a\xi - 1 = 0$.

The intersections by the arbitrary line are $ux + b a y = 0$ which give the points corresponding to the intersection of $\xi + \xi = \xi + \xi$ by the quadric $\xi = \xi, a\xi - 1 = 0$.

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giving four intersections. But the intersections by the line $by + cu = 0$ that is by any line through the point $y = 0, x = 0$ are obtained from the equation $A\xi^2 + C\xi^2 = 0$, viz. this breaks up into $b\xi + b\xi = 0, \xi = 0$, and the last factor combined with the equation of the circle gives $b = 0, \eta = 0, \xi = b, \eta = -b$; the two circular points at infinity, corresponding each to the point $y = 0, x = 0$; the other factor gives four points corresponding to two variable points on the curve; viz. the line $by + cu = 0$ meets the curve in the point $y = 0, x = 0$ four times each at infinity in the curve $\xi + b\xi = 0$. The equations then becomes the point being like as the line at infinity in $\xi + \xi$ is, the line $by + cu = 0$ corresponding to the point being at infinity in $\xi + \xi$. We thus have for instance the conic ξ, η at the point at infinity is a node. The curve being trinodal (counting as two nodes) and one other node. The curve being trinodal (counting as two nodes) is of course one other node.

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On Geodesic Lines, in particular those of a Quadric Surface.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 191–211.
Read December 12, 1872.]*

THE PRESENT Memoir contains an investigation of the differential equation of the second order of the geodesic line on a surface, the coordinates of a point on the surface being expressed as functions of two parameters p, q ; and remarks on the connexion therewith; a deduction of Jacobi's differential equation of the first order in the case of a quadric surface, the parameters p, q being those which determine the two sets of curves of curvature; herein when the parameters are those which determine the two right lines through the surface; and a discussion of the forms of the geodesic lines in the two cases of an ellipsoid and a sheet hyperboloid respectively.

Preliminary Formulæ.

1. I call to mind the fundamental formulæ in the Memoir by Gauss, “Disquisitiones generales circa superficies curvas,” *Comm. Gött. recent.*, t. VI. 1827, (reprinted as an Appendix in Liouville's edition of Monge); together with some that I have added to them. The coordinates of a point on a surface are regarded as given functions of two parameters p, q ; these expressions of x, y, z thus determining the equation of the surface, and we have

$$\begin{aligned} dx &= \frac{1}{2} dx \cdot a_1 + \frac{1}{2} dy \cdot a_2 + \frac{1}{2} dz \cdot a_3 + 2x da_1 + 2x dy \cdot a_2 + 2x da_3, \\ dy &= \frac{1}{2} dx \cdot b_1 + \frac{1}{2} dy \cdot b_2 + \frac{1}{2} dz \cdot b_3 + 2y db_1 + 2x dy \cdot b_2 + 2y db_3, \\ dz &= \frac{1}{2} dx \cdot c_1 + \frac{1}{2} dy \cdot c_2 + \frac{1}{2} dz \cdot c_3 + 2y dc_1 + 2x dy \cdot c_2 + 2y dc_3; \\ A, B, C &= bc - cb, ca - ac, ab - ba. \end{aligned}$$

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where, differential equation of surface is

$$A dx + B dy + C dz = 0.$$

Also

$$E, F, G = a^2 + b^2 + c^2, \quad aa + bb + cc, \quad a^2 + b^2 + c^2;$$

so that element of length on the surface is given by

$$ds^2 = dp^2 + dq^2 + (E dp + F dq + G dq^2).$$

The equation $(E, G, (dF \cdot dq)^2) = 0$ determines at each point on the surface two directions (commonly imaginary) which are called the “circular” directions. Passing on the surface in any direction from the point in question, the directions of this current (always imaginary) which are the “circular” curves; the equation $(E, F, G, (dp, dq)^2) = 0$ is the differential equation of these curves; and if we have $E = 0, G = 0, E, dq = G dp^2 + 0$; because the differential equation of the circular curves are geodesic lines.

I write also

$$E^a, F^a, G^a = aA + bB + cC, \quad aA + bB + cC, \quad a^aA + b^aB + c^aC,$$

or, what is the same thing, $\{aa^a, (BA, B, A)\}, \{aa^a, (b^aB, b^aC)\}, \{aa^a, (c^aA, c^aB)\}$, respectively,

$$\begin{vmatrix} a, & b, & c \\ a, & b, & c \\ A_a, & A_b, & A_c \end{vmatrix}, \begin{vmatrix} a, & b, & c \\ A, & B, & C \\ A, & B, & C \end{vmatrix}, \begin{vmatrix} a, & b, & c \\ A, & B, & C \\ A^a, & B^a, & C^a \end{vmatrix},$$

(those last symbols do not occur in Gauss). They are the D, D, D^a, D^a of Liouville, 2. The radius of curvature of normal section corresponding to the direction $dp : dq$ is given by

$$P = \frac{(E, F, G, (dp, dq))}{\sqrt{(E^a + 2F dp dq + G^a dq^2)}};$$

whence it appears that the directions of the inflexional or chief tangents (or Hess- *tangents*) are determined by the equation

$$(E^a, F^a, G(dp, dq)) = 0.$$

The directions in question are imaginary on a surface like the ellipsoid (then they are the same as the same one of two curves form the chief curves); but at any geodesic lines. We say on the surface even one of two curves form the chief curve; the chief curves they are at any geodesic line and are the chief curves.

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the differential equation of these is $(E^a, F^a, G(dp, dq)) = 0$, and in particular if $E^a = 0, G^a = 0$, then this becomes $dp dq = 0$, we have $p = \text{const.}, q = \text{const.}$ for the equations of the two sets of chief curves respectively. On the hyperboloid the chief curves are the two sets of generating lines. The chief curves are not in general geodesic lines, but on the hyperboloid, and though lines, they are, it is clear, geodesic lines.

3. The directions of the curves of curvature, or the principal tangents, and the corresponding radii of the radii of curvature are determined by

$$P = \frac{E dp + F dq}{E^a dp + F^a dq} = \frac{F dp + G dq}{F^a dp + G^a dq};$$

or, what is the same thing, these directions are determined by the equation

$$\begin{vmatrix} dp^2, & -dp dq, & dq^2 \\ E, & F, & G \\ E^a, & F^a, & G^a \end{vmatrix} = 0.$$

The same equations may be written

$$-dp : dp : dq = \frac{aE - F^a}{F^a - F^a F} = \frac{F^a - FG}{F^a - FG} = \frac{F^a - FG}{FG^a - F^a G};$$

that is the principal radii of curvature are determined by the equation

$$P^2(EG - F^a) - P(EG^a + E^a G - 2FF^a) + (E^a G^a - F^{a^2}) = 0;$$

(last term is $= P^a$, but it is better to retain the original form); and then, ρ being either root, the last preceding equation gives the direction of the curve of curvature corresponding to the given value of the radius of curvature.

If $p = \text{const.}, q = \text{const.}$ are the equations of the two systems of curves of curvature respectively, then the quadric equation in (dp, dq) must reduce itself to $dp dq = 0$, viz. we have $F = 0, F^a = 0$; and the principal equations among these the more simple form; in particular, in this case, instead of forming the equation in (dp, dq) , we have for the principal radii of curvature

$$\rho^a + p = \frac{E}{F^a}, \quad \rho^a + p = \frac{G}{G^a}.$$

If $p = \text{const.}, q = \text{const.}$ are the equations of the two systems of curves of curvature respectively, then expressing that the curve which passes through the generating lines. We may as before observe the lines $p = \text{const.}, q = \text{const.}$ of the chief lines, but in this case, as before, the equation $F = 0, F^a = 0$ and the generating equations.

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General Theory of the Geodesic Lines on a Surface.

4. I now proceed to investigate the theory of geodesic lines on a surface, the surface being determined as above by means of given expressions of the coordinates x, y, z in terms of the parameters p, q .

The differential equation obtained by Gauss for the geodesic lines is in a form not symmetrical in regard to the two variables; viz., the equation is in a form

$$\frac{dE}{dp} dq + 2 dE \frac{dF}{dq} dp + \frac{dG}{dq} d^a q = 2 ds \frac{E dp + F dq}{ds};$$

where, as above,

$$ds = (E, F, G(dp, dq)).$$

If we herein consider p, q as functions of a variable parameter θ , and we further not symmetrical equation in p, q , viz.

$$dp d^a q - dq d^a p, dq^2, dp dq, dp^a = 2 ds (F, F, G(dp, dq)) = 0,$$

$$dq^2, E, F, G, d^a p, d^a q;$$

also

$$H_a + F = E, G, d^a q, \&c.,$$

then the equation is

$$(E, F, G dp, dq) = \Sigma_a (H(F dp + F dq) - E(F dp + G dq)) d^a q, \&c..$$

We have

$$\left(\frac{F' dp + F' dq}{\sqrt{ds}} \right)^a = \frac{1}{ds} (M + N),$$

where X is the part containing p, p^a , and only the first differential coefficients of p^a no second differential coefficients of p ; and N is the second differential coefficients of p , viz., for the simply, in regard to the parameter θ , $E = E dp$, and adding, we have

$$X = -\frac{1}{2} p^a (R dp + F' dq) - \frac{1}{2} p^a F' dp + F' dq + Q (F dq + F' dp) dq + dq (R (F dp + F' dq)) + E dp + F dp,$$

or substituting for Ω its value, this becomes

$$= -p^a (E^a F' + Q EF + F F') - p^a (EFF' - E F' G^a) - p^a F' dp G + p q (F F' G - F F' G) - EF dp + F F' dq;$$

and the second term is contained, and the curve is then in evidence as follows:

$$\frac{dE}{dp} dp + \frac{dE}{dq} dq + \frac{dE}{dq} (ER - EF) (dp + Q) (EFEF + 0) dq, + 2EF (dp dq + dq E dp + F dq) dq dp (R - Q) + (Q (EF - E$$

where we know that the whole equation must divide by q .

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5. We have

$$M = (\nu E + \alpha)(E dp + F dq)^a + (E dp + F dq)(EFq + 4yF^a)(F dp + G dq) + p^a (EE + 2FF' - FG)(F dq + G dp) + 4F' (I$$

or say

$$2M = (-p^a + Fp + 4yF^a)X + 4F'(2F + G)(F + G) + 4YBY;$$

where, putting for shortness $X, Y, Z = E dp + F dq, F dp + G dq, EF'p + G dq$, etc., respectively (these are functions of p, q, p^a, q^a only),

$$\begin{aligned}\beta &= \frac{dE}{dp} - 2\frac{dF}{dq} + 4p^a, & p^a &= \frac{dE}{dp}, \\ \gamma &= -2\frac{dE}{dp} + 4p^a \frac{dF}{dq} + 3p^a; \\ \delta &= +2\frac{dE}{dp} - 4p^a \frac{dF}{dq} + 4p^a, & p^a &= \frac{dE}{dq}; \\ \epsilon &= +\frac{dG}{dp} - 4p^a \frac{dG}{dq} + p^a + 3p^a; \\ B &= \frac{dG}{dp} + 4p^a \frac{dG}{dq} + 4p^a, & p^a &= \frac{dE}{dq}.\end{aligned}$$

The form in g^a in EE^a , which is in fact $= a^a (Z(p^a + Eq)^a - 2Y(p^a q^a + Z(dp + Eq^a)^a)$. Presenting now the value of X , the geodesic equation thus becomes

$$-p^a (EE^a + p^a (EE^a + p^a (EFE^a - EE^a) - p^a FPG E + 2FPG E) - E^a F' dq E^a + F dq E^a + (-p^a (EE^a + p^a (p^a + 4yF^a)$$

which is the required differential equation of the second order; we shall see, however, that this is in another form, viz. as a more simple equation between p^a and q^a , supposing p, q to be functions of the parameters θ ; not, as in the conjugate substitution).

The condition in question is expressed by the equation:

$$\begin{vmatrix} A, & B, & C \\ xy + dpy xy + dyy, & y^a + y dy y + y dqy, & xy \\ xy + dqy xy + dyy, & y^a + dy y + y dqy, & y dqy \end{vmatrix} = 0,$$

that is

$$\begin{vmatrix} A, & B, & C \\ ay^a + dy, & by^a + dy, & cy^a + cyy \\ ay^a + ay, & by^a + by, & cy^a + cyy \end{vmatrix} = 0.$$

9. The first determinant in the sum of two terms each as $A(bc^a)B(yz - zy)$, etc., $(A(yca^a) - B(2y - xy) + (xD - F(2y) - p^a Fp^a)$; The second determinant is the sum of three terms each as $a(yca^a - zy)$, etc., where the factor $\{, \}$;

$$\rho^a [a(b(bc - cb(bc(a^a) + c(b(bc - c(b(bc(a^a) (b(bc(a^a) (b(bc(a^a) (b(bc(a^a) ($$

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7. Instead of starting, as above, from the equation given by Gauss, we may use the geometrical property that at each point of a geodesic line the osculating plane passes through the normal of the surface.

Considering, as above, p, q as functions of a variable θ , then θ becoming $\theta + d\theta$, the new values of p, q are

$$p + p^a d\theta + \frac{1}{2} p^a d\theta^2, \quad q + q^a d\theta + \frac{1}{2} q^a d\theta^2,$$

and then those of x, y, z are

$$x = ap^a d\theta + bq^a d\theta + \frac{1}{2} d\theta^2 (ap^a + bq^a + 2a p^a q^a + 2b p^a q^a + 2b p^a q^a),$$

$$y = bp^a d\theta + bq^a d\theta + \frac{1}{2} d\theta^2 (bp^a + bq^a + 2a p^a q^a + 2b p^a q^a + 2c p^a q^a),$$

$$z = cp^a d\theta + cq^a d\theta + \frac{1}{2} d\theta^2 (cp^a + cq^a + 2a p^a q^a + 2b p^a q^a + 2c p^a q^a),$$

The condition in question is expressed by the equation

$$\begin{vmatrix} A, & B, & C \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{vmatrix} = 0,$$

that is

$$\begin{vmatrix} A, & B, & C \\ ap^a + bq^a, & bp^a + bq^a, & cp^a + cq^a + cy^a \\ ap^a + aq^a, & bp^a + bq^a, & cp^a + cq^a \end{vmatrix} = 0.$$

8. The first determinant is the sum of three terms such as $A(bc^a - cb^a)p^a$, &c., viz. this is $A(p^a y^a - q^a y^a)$, where A stands for $AD - BC$ (viz. $\equiv FF^a - EF^a$).

The second determinant is the sum of three terms such as $a p^a (bc^a - cb^a)$, &c., where the factor $\{, \}$ is

$$p^a [a (b (bc - cb (bc^a)) + c (b (bc - c (b (bc^a)) (b (bc^a)) (b (bc^a)) (b (bc^a))];$$

The second determination is $-2p^a q^a (EE^a + (2D - p^a + Pp^a) + p^a (Pq^a + p^a + q^a))$ viz. taking $B = u^a EE^a + 2EF^a y^a + Gq^a$. $((PD - p^{aa}) + p^a Pp^a + p^a)$.

The second determination is $-2p^a q^a (EE^a + (2D - p^a + Pp^a) + p^a (Pq^a + p^a + q^a))$ where the factor $\{, \}$ is

$$p^a [Y(a a - c)(a^a)] + 2p^a q^a [Z(a^a + a^a)(b^a) + a^a + q^a (R(a^a) + p^a + q^a(b^a))];$$

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which is the second determinant; the second factor is, as above $A((py^a + q^a + R(p^a + q^a))$. We have thus the second determinant,

$$(EX - F^a)p^a + 2(FQ + FF^a)p^a q^a + 2(FF^a + GQ)p^a + R(p^a + q^a) = 4F^a(2p q + Q p^a + Q p^a) - p^a(aB^a + BY' EE^a p^a +$$

an equation of the same form as that previously obtained, and which has been differently identified therewith.

The Circular Curves are Geodesics.

9. I proceed to show that the circular curves are geodesics; viz. that an integral of the geodesic equation is

$$(E, F, G)(\xi, \eta)^2 = 0.$$

Starting from this equation, we have

$$2(E\xi + F\eta)(E\xi + F\eta) + 2(F\xi + G\eta)(F\xi + G\eta) + (E^a\xi^a + 2F^a\xi\eta + G^a\eta^2)(\xi + \eta)d\xi + (E^a\xi^a + 2F^a\xi\eta + G^a\eta^2)(\xi + \eta)d\eta =$$

Now the equation, writing therein $E\xi^a + Fq = q^aw$, gives $Fq + Qpq = q^aw$; their expansion may be written

$$4F\eta(E\xi + F\eta) = 0;$$

the value of w being therefore $-p^a - 2Bp^a + q^a$. The result then divided thus becomes $2Bw(E\xi - F\eta)(Fq + G\eta^a) + (E\xi G - E\xi G^a)(\xi w^a + 2F^a\xi\eta + G^a\eta^2) = 0$,

$$= \Sigma(\xi + Zw)(F^a + Gw^a),$$

that is

$$2(E(E\xi + F\eta)(2Eq - F^a) + (E\xi G + 2(F^a F^a + F^a + G^a w^a)2w) + (ay + bw)(F^a + bw^a) - (aw + b\xi + dw)(Pu + 2F\xi y + dw^a$$

viz. taking $\beta E\xi = (Q Q^a + F^a w + d^a)$; this agrees with the geodesic equation.

The foregoing integral, $(E, F, G)(\xi, \eta)^a = 0$, I believe, a particular, not a singular, solution of the geodesic equation.

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The Chief Lines are not in general Geodesics.

10. That the chief lines are not in general geodesics appears most readily as follows:

To find the condition in order that $p = \text{const.}$ may be a geodesic, we write in the geodesic equation $p^a = 0$, $p^a = 0$; the equation thus becomes

$$F^a q^a - Bq^a + EE^a q^a = 0,$$

that is to say

$$2EE^a = E^a B - EB^a,$$

as the condition in order that $p = \text{const.}$ may be a geodesic; the condition that it may be a chief curve is $E^a = 0$, which is a different condition.

We have of course, in like manner,

$$2GG^a = G^a B - GB^a,$$

as the condition in order that $q = \text{const.}$ may be a geodesic; the condition that it may be a chief curve is $G^a = 0$. If, in any case, both conditions are satisfied, we will have

$$FG - F^a G' + GE^a = 0, \quad EF^a - EE^a F + EE^a = 0,$$

and that is satisfied. If $p = \text{const.}$ and $q = \text{const.}$ are both of them sets of lines on the surface, then the four equations are satisfied.

Special Form of the Geodesic Equation.

11. In the case where the curves $p = \text{const.}$, $q = \text{const.}$ intersect at right angles (and in particular when they are the curves of curvature), we have $F = 0$, then since also $F^a = 0$, $F^a = 0$; and the geodesic equation becomes one much more simple form,

$$2(E\xi^a + Qp)(E\xi G - Gp^a)(pE^a + p^a + q^a + q^a) + (E\xi G - F^a G^a)q^a + 2(E\xi G^a - F^a G^a)q^a q^a;$$

[E^a] In the case of a surface of revolution we have

$$dx = (1 + F^a)dp^a + p dq^a,$$

that is

$$\begin{aligned} E &= 1 + F^a, & E^a &= TP^a, & EE^a &= 0, \\ G &= p^a, & G^a &= 0, & G_a^a &= 0; \end{aligned}$$

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hence the differential equation is

$$(1 + F^a)p^a(q^a - p^a)q^a - p(d(p)^a + p)^a q^a = 0;$$

this has an integral

$$\frac{(1 + F^a)p^a}{p^a} + \frac{1}{p^a} = C,$$

or say

$$Cdp^a = p^a(p - C^a)dq^a;$$

where

$$Cp + Q = p^a + p^a;$$

Writing here ρ, ϕ for p, q , where ρ is the distance of the point from the axis, and ϕ is the longitude reckoned from an arbitrary meridian, then the equation is

$$Cdq = ax,$$

which is the equation given by Legendre, *Théorie des fonctions elliptiques*, t. 1, p. 361. This may also be written

$$\frac{C^a}{C} = \cos \psi$$

if ψ be the inclination of the geodesic line to the parallel of latitude.]

Geodesics on a Quadric Surface.

12. In the case of a quadric surface, $\frac{x^a}{a} + \frac{y^a}{b} + \frac{z^a}{c} = 1$; writing for shortness $x, y, z, =, e - a, -e - b, e - c$, and we may express the coordinates x, y, z in terms of two parameters p, q as follows:

$$x = \frac{2py(a+p)(a+q)}{(a-b)(a-c)}, \quad y = \frac{2py(b+p)(b+q)}{(a-b)(a-c)}, \quad z = \frac{p^a(a+p)(a+q)}{(a-b)(a-c)};$$

where, in fact, $p = \text{const.}, q = \text{const.}$ are the equations of the two sets of curves of curvature respectively. Writing moreover

$$P = \frac{1}{(a+p)(b+p)(c+p)}, \quad Q = \frac{Q}{(a+q)(b+q)(c+q)};$$

that is

$$dx = \frac{1}{2}(b-q)p^a P dp, \quad \frac{1}{2}(b-p)q^a Q dq;$$

and the geodesic equation becomes

$$E, F, G((p-q), 4(b-q)P, (b-q)Q, (b-q));$$

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where E^a , G^a stand for $\frac{de}{dq}$ and $\frac{de}{dq}$ respectively, viz. this is

$$\rho^2 = \frac{P^a}{Q^a} = P[q + (q dQ dp + P dQ dq)] + p^a y P + (q dq) dQ + Q dP + q + Q^a, = P[p - q + (q PP - P Q)Q] dq.$$

or, differentiating logarithmically, this gives

$$\frac{2p^a P^a}{p^a + Q^a} (p - q) + \frac{P^a}{q + Q^a} (p^a + Q^a) dq + \frac{P^a}{P^a} (p^a - q + Q^a) = 0;$$

viz. differentiating the denominators, this gives

$$\frac{pPP^a - p^a P^a P^a}{P^a + Q^a} - \frac{QdP^a + QdPd^a Q + (PP^a - P^a Q)(P^a - p^a Q + P^a P^a) dq}{p^a + Q^a + Q^a} = \frac{P^a}{P + Q^a} (p^a - Qd^a Q + (PP^a - P^a Q) dq)$$

that is

$$= R(p - p) (PP^a - P^a Q) dq + (PP^a - P^a Q) (p^a + Q^a + Q^a),$$

or

$$= R(p - p) Q + (P + Q^a) Q^a + Q^a P;$$

We have next that

$$(pPP^a - p^a P) (p^a - Q) - p^a (P^a + Q) (P^a - P^a Q + P^a) + p^a (qP - Q) (P^a - P^a Q);$$

which holds for any function f , we require its value when

$$f = -p a - b - c - p, 2(P - p)(P + p)Q;$$

and the differential equation, $(E, F, G)(\xi, \eta)^a = 0$, of the chief lines becomes

$$\xi^a (p a + p + p c + p) + (E + Q)(p^a + QP^a) d^a Q + (p a + Q)(P^a Q + P^a Q) (p^a + Q^a P) = 0,$$

involving the arbitrary constant θ as the differential equation of the first order of the geodesic lines on the quadric surface, $\frac{x^a}{a} + \frac{y^a}{b} + \frac{z^a}{c} = 1$; the geodesic lines in question

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all touch the curve of curvature determined by the parameter θ , that is the curve which is the intersection of the surface by the confocal

$$\frac{x^a}{a+\theta} + \frac{y^a}{b+\theta} + \frac{z^a}{c+\theta} = 1.$$

13. In the particular case $\theta = \alpha$, the equation becomes

$$P dp^a - Q dq^a = 0,$$

that is

$$\frac{p dp^a}{p(p+p)(p+p) - p a(p+p)} = 0,$$

which is the differential equation of the circular curves on the surface.

14. The signification of the case $\theta = b$ is not at first sight so obvious. Supposing that θ is first indefinitely small, and writing the equation in the form

$$\frac{\xi^a}{p} + \frac{\eta^a}{q} + \frac{\xi^a}{c} + \frac{1}{p^a} = 0,$$

we have at once that the equation $F = 0$ is

$$\frac{x^a}{a} + \frac{y^a}{b} + \frac{1}{p} \left(1 - \frac{x^a}{a^a} \right),$$

$$\frac{1}{p} \eta^a - \frac{p^a}{p(p+p)} \left(1 - \frac{x^a}{a} \right)$$